

Advanced MLOps Exercise

Professional CI Workflow with Pull Requests, Docker & Container Registry

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Repository: `github.com/Mahad1090/student-ml-api` · **Registry:** `ghcr.io/mahad1090/student-ml-api`

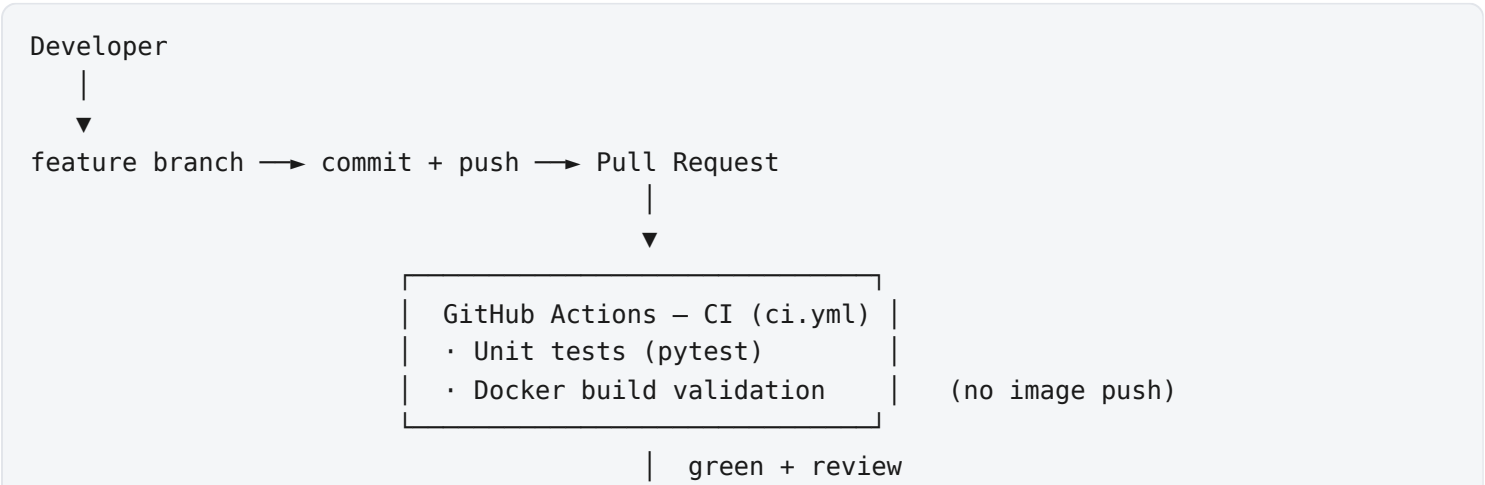
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Core principle. Git manages the evolution of source code. Pull Requests control how changes enter `main`. CI verifies those changes. Docker converts approved source code into a reproducible artifact. The container registry stores and distributes versioned artifacts that can later be delivered consistently to staging and production.

Table of Contents

1. Pipeline Overview	11. Container Registry (Parts 12, 16)
2. Requirement Completion Matrix	12. Git Tags & Release Workflow (Parts 13–15)
3. Application & Automated Tests (Parts 1–2)	13. Artifact Reproducibility (Part 17)
4. Git Workflow & Pull Requests (Parts 3–4)	14. Version 1.1.0 (Parts 18–19)
5. GitHub Actions CI (Part 5)	15. Rollback Exercise (Part 20)
6. Deliberate Failure Demonstration (Part 6)	16. Traceability Challenge (Part 21)
7. Branch Protection (Part 7)	17. CI vs Release, Metadata, Commit Tag, Cache (Parts 22–25)
8. Merge Strategy (Part 8)	18. Failure Analysis (Part 26)
9. Dockerization (Part 9)	19. Viva Questions
10. Local Build, Run & Inspection (Parts 10–11)	20. Screenshot Appendix

1. Pipeline Overview



▼
Squash merge → main

↓
▼
Semantic version tag vX.Y.Z

↓
▼
| GitHub Actions – Release (release.yml) |
| · pytest · GHCR login |
| · docker build · derive version |
| · push X.Y.Z / latest / <sha> |
↓
▼

ghcr.io/mahad1090/student-ml-api
|— 1.0.0 |— 1.1.0 |— latest

2. Requirement Completion Matrix

Part	Requirement	Status	Evidence
1	Flask/FastAPI app with <code>/health</code> + <code>/predict</code>	Done	§3, Fig 14
2	≥ 4 automated tests, <code>pytest</code> green	Done	§3, Fig 1
3	Feature-branch workflow, meaningful commits	Done	§4, Fig 3, 17
4	Professional PR description + checklist	Done	§4, Fig 2, 18
5	CI on PR: checkout → Python → deps → tests → docker build	Done	§5, Fig 6, 19
6	Deliberate test failure + fix, evidence captured	Done	§6, Fig 5, 6, 3
7	Branch protection on <code>main</code> , documented	Done	§7, Fig 7
8	Merge PR (strategy stated + justified)	Done — Squash	§8, Fig 4
9	Production Dockerfile + <code>.dockerignore</code>	Done	§9
10	Local build <code>student-ml-api:1.0.0</code> , run, verify	Done	§10, Fig 12–14
11	images / ps / logs / inspect / exec; identify IDs	Done	§10, Fig 15
12	Container registry (GHCR)	Done	§11, Fig 8
13	Git tag <code>v1.0.0</code> after merge	Done	§12, Fig 17
14–15	Tag-triggered release workflow, version derived from tag	Done	§12, Fig 10, 11

16	Registry contains <code>1.0.0</code> , <code>1.1.0</code> , <code>latest</code> ; digest recorded	Done	§11, Fig 8, 9, 21
17	Delete local image, pull from registry, run	Done	§13, Fig 16
18–19	v1.1.0 via full PR workflow, released	Done	§14, Fig 18–20, 11
20	Rollback to <code>1.0.0</code> from registry, no rebuild	Done	§15, Fig 21, 23
21	Full traceability chain for 1.1.0	Done	§16
22	CI vs Release separation + rationale	Done	§17
23	OCI labels on image	Done	§17, Fig 15
24	Commit-SHA image tag	Done	§17, Fig 8
25	Docker build-cache analysis	Done	§17
26	≥ 2 failures reproduced & diagnosed	Done — 3	§18, Fig 5, 25

3. Application & Automated Tests (Parts 1–2)

The service is a **FastAPI** application (`app.py`). The prediction "model" simply doubles the input — the exercise targets the MLOps workflow, not model quality.

Endpoint	Request	Response (v1.0.0)
GET <code>/health</code>	—	<code>{"status":"healthy","application":"student-ml-api","version":"1.0.0"}</code>
POST <code>/predict</code>	<code>{"value":10}</code>	<code>{"input":10,"prediction":20}</code>

The application version is read from the `VERSION` file at runtime, so it is never hard-coded in code. Pydantic `StrictInt` | `StrictFloat` validation makes a non-numeric `value` return HTTP 422 rather than being silently coerced.

Test suite — `tests/test_app.py` (4 tests)

- `test_health_endpoint` — status, application name, version
- `test_predict_success` — `{"value":10}` → `{"input":10,"prediction":20}`
- `test_predict_missing_input` — empty body → 422
- `test_predict_invalid_input` — `{"value":"not-a-number"}` → 422

```
mahad1090@mahad-1090:~/Desktop/student-ml-api$ cd ~/Desktop/student-ml-api
.venv/bin/pytest -q
****
===== warnings summary =====
.venv/lib/python3.12/site-packages/starlette/testclient.py:40
/home/mahad1090/Desktop/student-ml-api/.venv/lib/python3.12/site-packages/starlette/testclient.py:40: DeprecationWarning: The anyio.abc.BlockingPortal alias is deprecated, use anyio.from_thread.BlockingPortal instead.
PortalFactoryType = typing.Callable[[], typing.ContextManager[anyio.abc.BlockingPortal]]

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
4 passed, 1 warning in 1.15s
mahad1090@mahad-1090:~/Desktop/student-ml-api$
```

Fig 1 — `pytest -q` → 4 passed.

4. Git Workflow & Pull Requests (Parts 3–4)

Direct development on `main` was never performed. All code reached `main` through three Pull Requests:

PR	Branch	Title	Merge commit
#1	feature/prediction-api	feat: prediction API with CI pipeline	fa0394b
#2	feature/model-metadata	feat: report application and model version on /health	02dc3db
#3	docs/mlops-workflow-writeup	docs: MLOps workflow write-up, traceability and failure analysis	4650a2b

Commits follow Conventional-Commits style: `feat:`, `test:`, `ci:`, `fix:`, `docs:`, `chore:`. Each PR body contains **Summary** · **Changes** · **Testing Performed** · **Docker Impact** · **Checklist** (rendered from `.github/pull_request_template.md`).

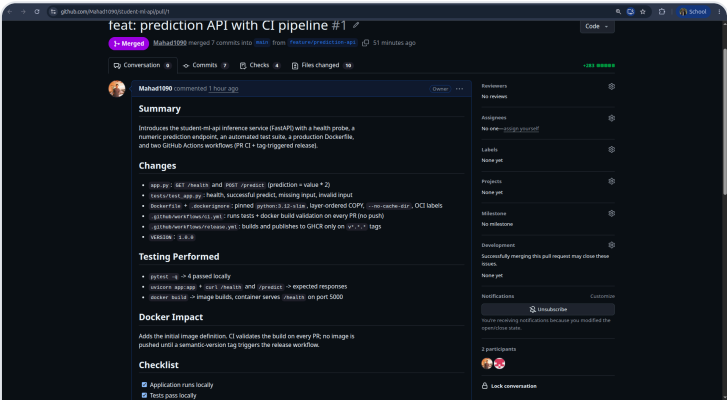


Fig 2 — PR #1 professional description & checklist.

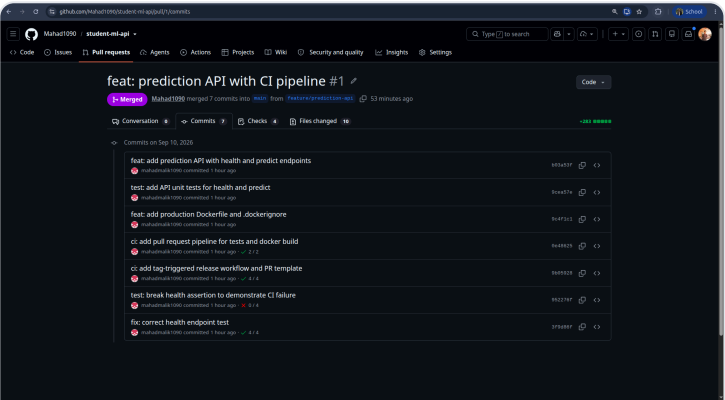


Fig 3 — PR #1 commit history (7 commits incl. the deliberate **break** X and the **fix** ✓).

5. GitHub Actions CI (Part 5)

`.github/workflows/ci.yml` runs on `pull_request` → `main` and on pushes to feature branches. It publishes **no** artifact.

```
job: test          → checkout → setup-python 3.12 → pip install -r requirements.txt → pytest -q
job: docker-build  → needs: test → checkout → docker build (validation only, no push)
```

Because `docker-build` declares `needs: test`, a failing `pytest` short-circuits the pipeline (the build job is skipped) and the whole run is **FAILED**. A failing `docker build` likewise fails the run. Branch protection then makes that a hard merge gate.

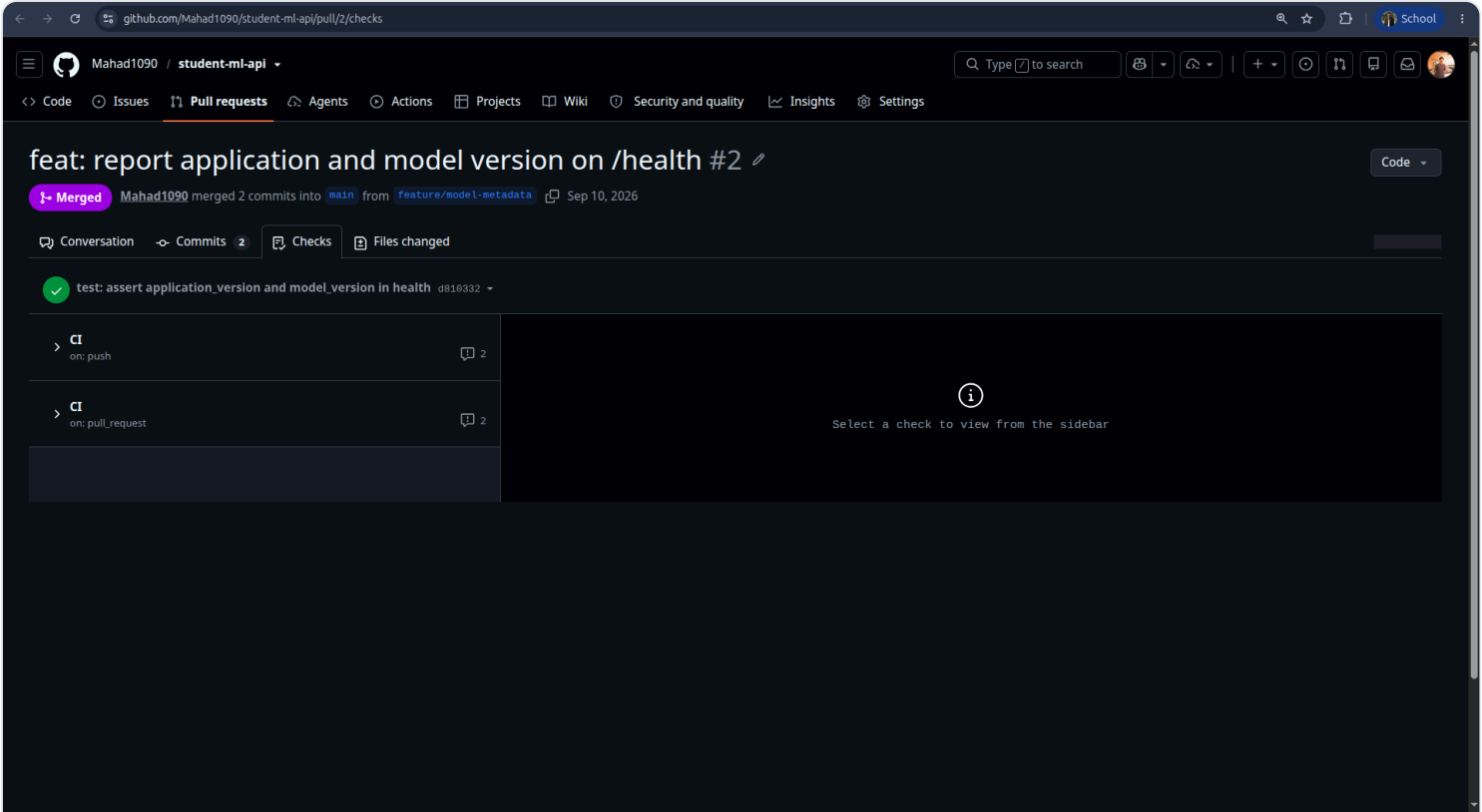


Fig 19 — required checks (Unit tests , Docker build validation) on a PR.

6. Deliberate Failure Demonstration (Part 6)

	Detail
Symptom	PR #1 CI turned red . Unit tests failed; Docker build validation was skipped; PR banner "Merging is blocked".
Root cause	Commit 952276f changed test_health_endpoint to assert data["status"] == "wrong", but the endpoint returns "healthy".
Evidence	Actions run 34494638490 — E AssertionError: assert 'healthy' == 'wrong' → FAILED tests/test_app.py::test_health_endpoint (Fig 5).
Correction	Commit 3f9d86f "fix: correct health endpoint test" restored == "healthy". Re-run 34494921761 passed both checks; merge unblocked (Fig 6).

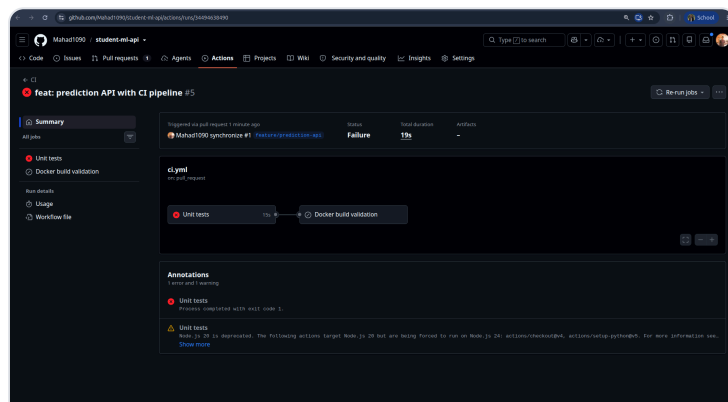


Fig 5 — CI run **FAILED** after the broken assertion was pushed.

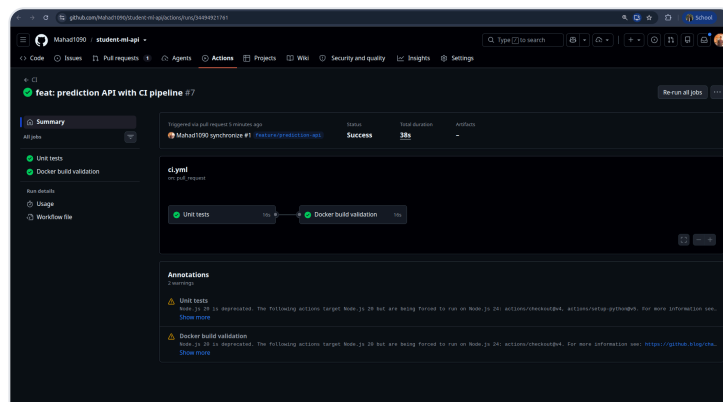


Fig 6 — CI run **SUCCESS** after the fix commit.

7. Branch Protection (Part 7)

Configured on `main`. Settings selected and their rationale:

Setting	Value	Why
Require a pull request before merging	On	No direct commits to <code>main</code> ; every change is reviewed and CI-checked.
Require approvals	1	Forces a second reviewer in a team context.
Dismiss stale approvals on new commits	On	An approval must reflect the code that actually merges.
Require status checks to pass	On — <code>Unit tests</code> , <code>Docker build validation</code>	Only a green pipeline can merge.
Require branches up to date before merging	On	Checks run against the real post-merge state.
Require conversation resolution	On	No unresolved review threads slip through.
Allow force pushes / deletions	Off	<code>main</code> history cannot be rewritten or the branch deleted.
Enforce for administrators	Off (solo repo)	GitHub forbids approving one's own PR; in a team this is <i>On</i> and a colleague approves. The PR + passing-CI gate still applies.

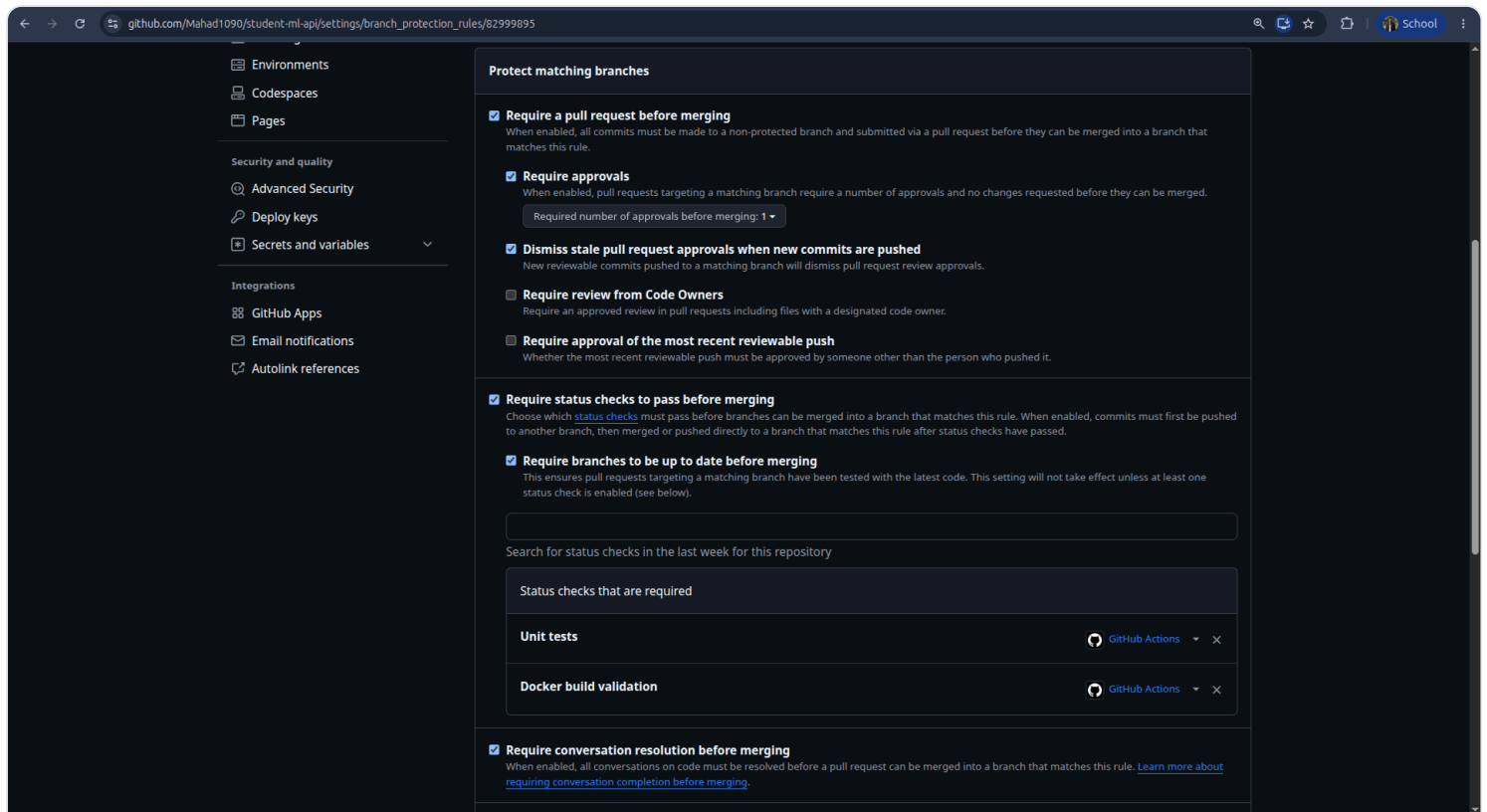


Fig 7 — branch protection rule for `main` .

8. Merge Strategy (Part 8)

Squash and merge was used for every PR.

- Each feature branch carries several iterative WIP commits (PR #1 even contains the deliberate break + fix). Squashing lands **one atomic, reviewed change** per feature on `main` .
- `main` stays linear and every commit is independently buildable, keeping `git bisect` meaningful.
- The squash commit references the PR number (`#1`) , (`#2`) — `main` → PR is one hop — while the full commit-by-commit history remains on the PR.

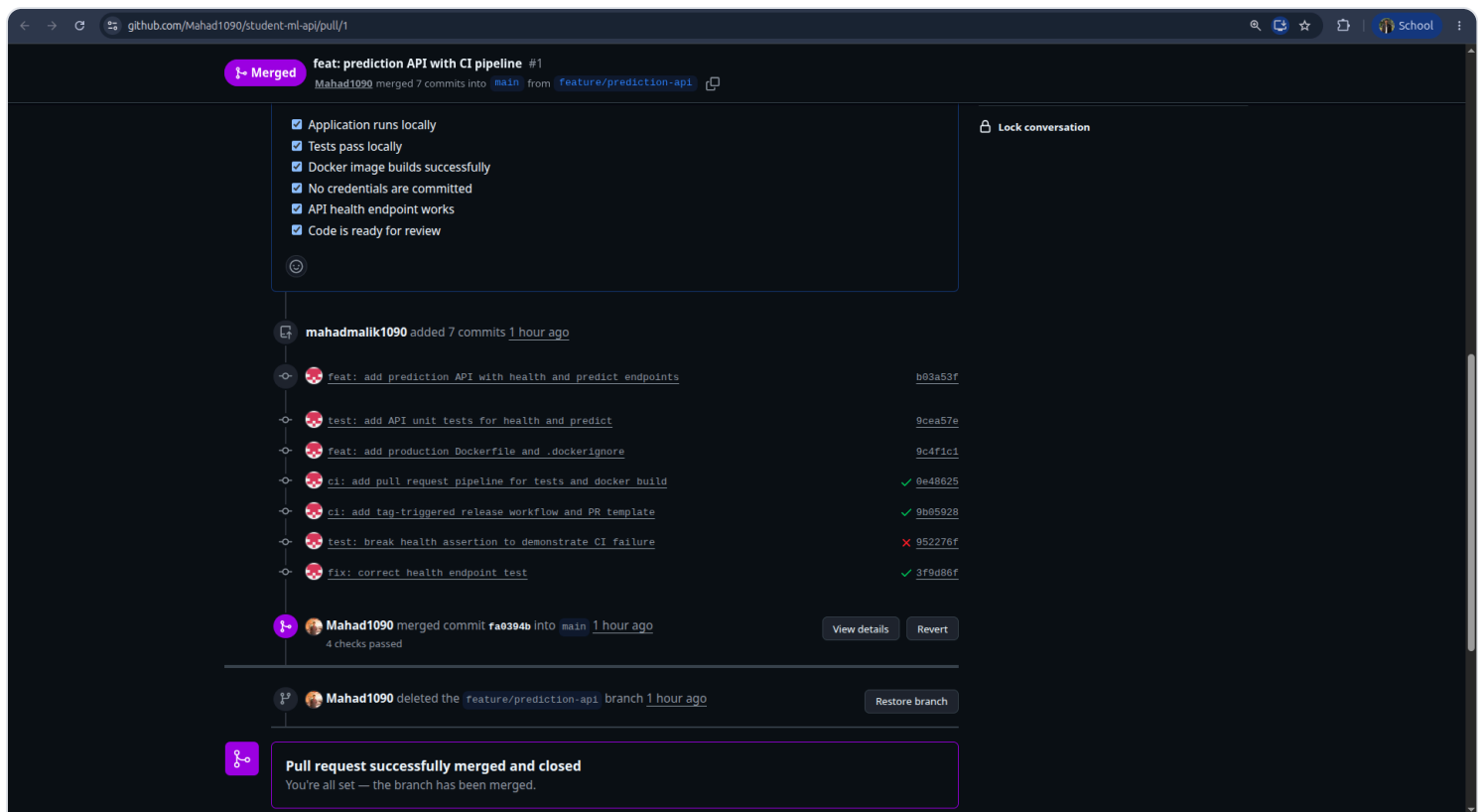


Fig 4 — PR #1 squash-merged as commit `fa0394b` ; checklist complete; branch auto-deleted.

9. Dockerization (Part 9)

```
FROM python:3.12-slim          # pinned – never python:latest
ARG APP_VERSION / GIT_COMMIT / BUILD_DATE
LABEL org.opencontainers.image.* # OCI metadata (Part 23)
WORKDIR /app
COPY requirements.txt .         # deps copied first ...
RUN pip install --no-cache-dir -r requirements.txt # ... so this layer caches
COPY app.py VERSION ./         # app code last
EXPOSE 5000
CMD ["uvicorn", "app:app", "--host", "0.0.0.0", "--port", "5000"]
```

`.dockerignore` excludes `.git`, `.github`, `__pycache__`, `*.pyc`, `.venv`, `.env`, `tests`, `*.md`, the assignment PDF, etc. Good practices demonstrated: explicit base-image version, `WORKDIR`, dependency-before-code `COPY` ordering, `--no-cache-dir`, `EXPOSE`, and an exec-form `CMD` that binds `0.0.0.0` (not `127.0.0.1`) so the port is reachable from the host.

10. Local Build, Run & Inspection (Parts 10–11)

```
docker build -t student-ml-api:1.0.0 \
  --build-arg APP_VERSION=1.0.0 \
  --build-arg GIT_COMMIT=$(git rev-parse HEAD) \
```

(Host port 5005 was used locally because ports 5000–5002 were occupied by unrelated containers — see §18 Failure 3.)

Identifier (Part 11)	Value
Container ID	cd0c8599377d
Image ID	b9bab104a906 (sha256:b9bab104a906...)
Exposed port	5000/tcp
Running command	uvicorn app:app --host 0.0.0.0 --port 5000
Application working directory	/app

Fig 12 — docker build steps 1–8.

Fig 13 — steps 9–11, **Successfully built / tagged.**

Fig 14 — container running from the local image; `/health` & `/predict` verified.

```
Fig 15— images / ps / logs / inspect / exec
+ OCI labels( revision = fa0394b678... ).
```

11. Container Registry (Parts 12, 16)

Registry: **GitHub Container Registry (GHCR)** — `ghcr.io/mahad1090/student-ml-api` , package visibility **Public**, linked to the repository.

Tag	Digest	Points to
1.0.0 , fa0394b	sha256:10000c57bd3adc45a796199743b8b52e858b54d996ca34433f208c6b7b778e4c	commit fa0394b
1.1.0 , latest , 02dc3db	sha256:2c9ccbd13d38ad9e8fbb4a067825f32b2c39b09082b31e4372f03332baadcaf5	commit 02dc3db

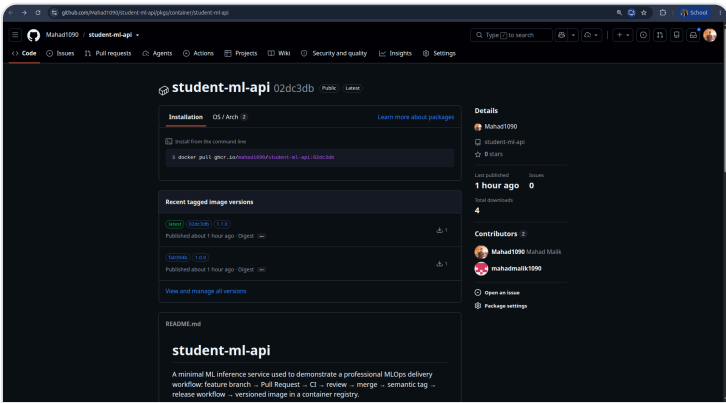


Fig 8 — GHCR package: tags latest , 1.1.0 , 1.0.0 + commit-SHA tags.

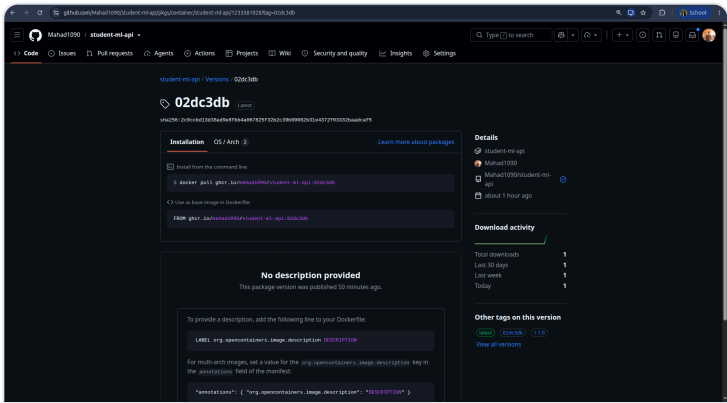


Fig 9 — version 1.1.0 / latest digest sha256:2c9ccbd1... .

12. Git Tags & Release Workflow (Parts 13–15)

After each merge: `git checkout main && git pull && git tag vX.Y.Z && git push origin vX.Y.Z` . Tags present: `v1.0.0 → fa0394b` , `v1.1.0 → 02dc3db` .

`.github/workflows/release.yml` runs **only** on `push: tags: ["v*.*.*"]` and performs:

checkout → derive version from tag → run pytest → GHCR login (GITHUB_TOKEN)
→ docker buildx build → tag X.Y.Z / latest / <short-sha> → push

The version is derived automatically — `version=${GITHUB_REF_NAME#v}` turns `v1.1.0` into `1.1.0` . Nothing is hard-coded; no password is stored in YAML (the GitHub-managed `GITHUB_TOKEN` with `packages: write` is used).

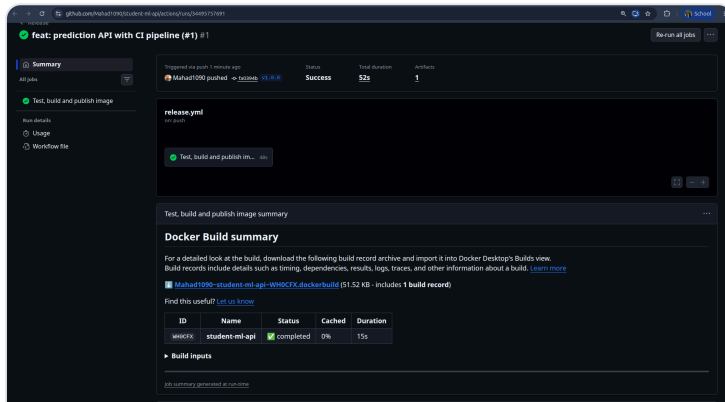


Fig 10 — Release run for v1.0.0 (success).

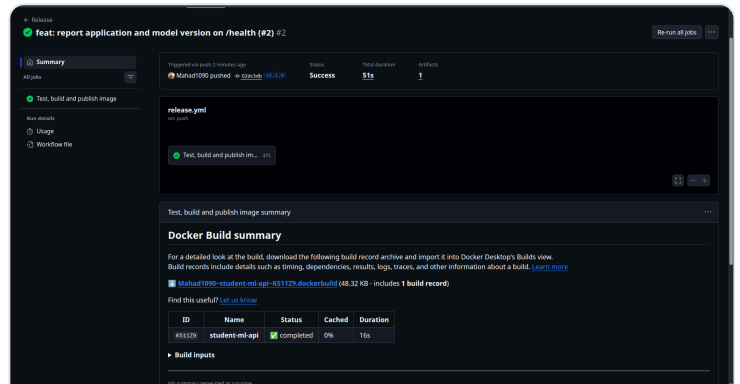


Fig 11 — Release run for v1.1.0 (success).

13. Artifact Reproducibility (Part 17)

```
docker rmi student-ml-api:1.0.0 # delete local build
docker pull ghcr.io/mahad1090/student-ml-api:1.0.0 # fetch from registry
  → Digest: sha256:10000c57bd3adc45a796199743b8b52e858b54d996ca34433f208c6b7b778e4c
docker run -d --name student-ml-api -p 5005:5000 ghcr.io/mahad1090/student-ml-api:1.0.0
curl localhost:5005/health → {"status":"healthy",...,"version":"1.0.0"}
```

The image ran on a different toolchain state **without rebuilding** — build machine → registry → runtime environment.

```

docker rmi student-ml-api:1.0.0

# pull the exact artifact the release workflow published
docker pull ghcr.io/mahad1090/student-ml-api:1.0.0
docker inspect --format '{{index .RepoDigests 0}}' ghcr.io/mahad1090/student-ml-api:1.0.0

# run the downloaded image
docker run -d --name student-ml-api -p 5005:5000 ghcr.io/mahad1090/student-ml-api:1.0.0
sleep 1
curl http://localhost:5005/health

WARNING! Your credentials are stored unencrypted in '/home/mahad1090/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
student-ml-api
Untagged: student-ml-api:1.0.0
Deleted: sha256:b9bab104a90690d3ad6432eb86a0407215784bbf364573ce7aeed49405b5b5b7
Deleted: sha256:530db45a1bdb456f08ba212cd2d2d17da97769e7f0d145c923b1a61816ff550
Deleted: sha256:4196da2ca7825f0d090b9adf92b5eb1a056cacc42542ca45b452f2e8c034096e
Deleted: sha256:0b152e3f699b1099a5336ccdb0bf6e308de2c11cd048704e95f753d0cf5066bc
Deleted: sha256:42abdbc694983f16f78bdcdb6863201c96329d815e8bceb746d65d726970064
Deleted: sha256:bc75e5f8e49cbdbcc25bbfcaf3ae0ef110f69f84014f6c3c74d45f1eb2a2b07
Deleted: sha256:9c5b1a872ee2928002d27c7e8b8ba6b29bc63fba484f3572629513592292d8b5
Deleted: sha256:83cc2ed46b357499ea61b37c4b117fd77afdd7501f17904f7b9181e4b6b22481
Deleted: sha256:be918425435e61488aeb31b4f3399054a83df9cbbcd1ee824e26eb41cee7b9b3
Deleted: sha256:85531d3bbd5c5eaa9098621d385dcdf7b749c8632cbb43f01845709d280639f2
Deleted: sha256:e788c7a508d6355c67e2af0d179e78330fb87a66f3e170548772b3fe13ab3256
Deleted: sha256:7b76cc7e13ea993a29ef731203c2938473a08063ac0004af2406f0c65f440f5d
Deleted: sha256:4b5bfc8de5dbb70e03f895598dc8b1b7b496b28d35aa1c61de8c6e7811873be1
Deleted: sha256:c7fe16b76e71d089731b7921e70fed364156abd5c7cd74059db57df1f27c5306
1.0.0: Pulling from mahad1090/student-ml-api
6310eb16bf42: Already exists
dbd0b7849e6c: Already exists
1560cfed01dc: Already exists
56235e424563: Already exists
5f9c87856cb7: Pull complete
a34f03daca35: Pull complete
650a66b0edce: Pull complete
7fa4cda34c04: Pull complete
Digest: sha256:10000c57bd3adc45a796199743b8b52e858b54d996ca34433f208c6b7b778e4c
Status: Downloaded newer image for ghcr.io/mahad1090/student-ml-api:1.0.0
ghcr.io/mahad1090/student-ml-api:1.0.0
ghcr.io/mahad1090/student-ml-api@sha256:10000c57bd3adc45a796199743b8b52e858b54d996ca34433f208c6b7b778e4c
42ffa09066c5bbe077e9afd40415156b76b9d8091842f4d1789378d0c42476bc
{"status":"healthy","application":"student-ml-api","application_version":"1.1.0","model_version":"model-1"}mahad1090@mahad-1090: ~/Desktop/student-ml-api$

```

Fig 16 — local image removed, pulled by digest from GHCR, container healthy.

14. Version 1.1.0 (Parts 18–19)

Branch `feature/model-metadata` . `/health` was extended to report the application and model versions independently, and `VERSION` bumped to `1.1.0` :

```

{"status":"healthy","application":"student-ml-api",
"application_version":"1.1.0","model_version":"model-1"}

```

Tests updated to assert both new fields. The full workflow was repeated (branch → commits → push → PR #2 → CI → merge → tag `v1.1.0` → release). The PR workflow was not bypassed.

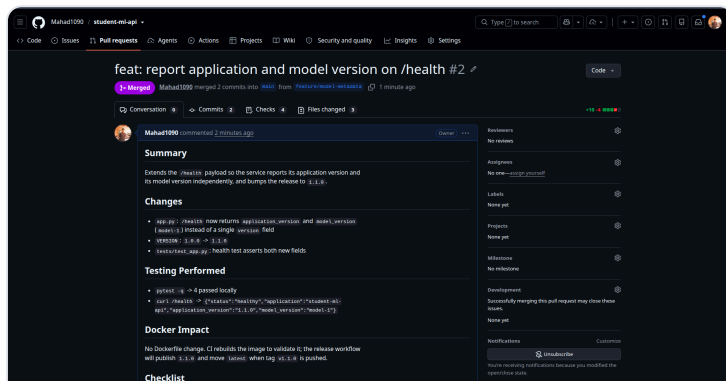


Fig 18 — PR #2 professional description.

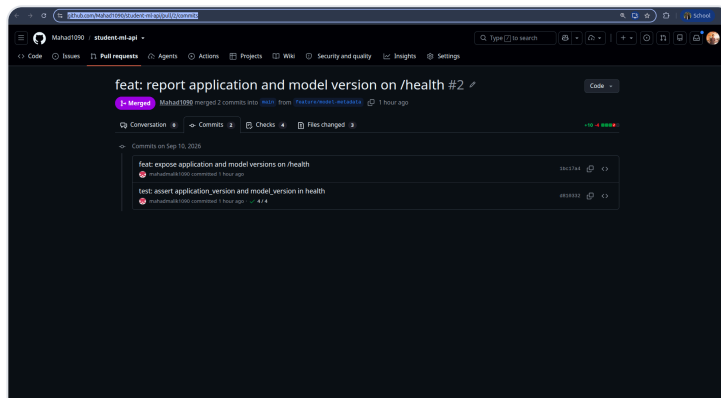


Fig 20 — PR #2 commits (feat + test).

```
mahad1090@mahad-1090:~/Desktop/student-ml-api$ docker rm -f student-ml-api
docker pull ghcr.io/mahad1090/student-ml-api:1.1.0
docker pull ghcr.io/mahad1090/student-ml-api:latest
docker pull ghcr.io/mahad1090/student-ml-api:1.0.0
docker images --digests | grep student-ml-api      # latest & 1.1.0 share a digest; 1.0.0 differs

# run 1.1.0
docker run -d --name student-ml-api -p 5005:5000 ghcr.io/mahad1090/student-ml-api:1.1.0
sleep 1; curl -s localhost:5005/health; echo      # application_version 1.1.0, model_version model-1

# Part 20 - ROLLBACK: no source change, no rebuild, just run the old artifact
docker rm -f student-ml-api
docker run -d --name student-ml-api -p 5005:5000 ghcr.io/mahad1090/student-ml-api:1.0.0
sleep 1; curl -s localhost:5005/health; echo      # back to version 1.0.0

student-ml-api
1.1.0: Pulling from mahad1090/student-ml-api
6310eb16bf42: Already exists
dbd0b7849e6c: Already exists
1560cfed01dc: Already exists
56235e424563: Already exists
9056e4fd24fd: Pull complete
49bec192185d: Pull complete
6143975ac7a8: Pull complete
6fcb0731b20e: Pull complete
Digest: sha256:2c9ccbd13d38ad9e8fbb4a067825f32b2c39b09082b31e4372f03332baadcaf5
Status: Downloaded newer image for ghcr.io/mahad1090/student-ml-api:1.1.0
ghcr.io/mahad1090/student-ml-api:1.1.0
latest: Pulling from mahad1090/student-ml-api
Digest: sha256:2c9ccbd13d38ad9e8fbb4a067825f32b2c39b09082b31e4372f03332baadcaf5
Status: Downloaded newer image for ghcr.io/mahad1090/student-ml-api:latest
ghcr.io/mahad1090/student-ml-api:latest
1.0.0: Pulling from mahad1090/student-ml-api
Digest: sha256:10000c57bd3adc45a796199743b8b52e858b54d996ca34433f208c6b7b778e4c
Status: Image is up to date for ghcr.io/mahad1090/student-ml-api:1.0.0
ghcr.io/mahad1090/student-ml-api:1.0.0
ghcr.io/mahad1090/student-ml-api 1.1.0 sha256:2c9ccbd13d38ad9e8fbb4a067825f32b2c39b09082b31e4372f03332baadcaf5 5d19dd07f99e 3 minutes ago 179MB
ghcr.io/mahad1090/student-ml-api latest sha256:2c9ccbd13d38ad9e8fbb4a067825f32b2c39b09082b31e4372f03332baadcaf5 5d19dd07f99e 3 minutes ago 179MB
ghcr.io/mahad1090/student-ml-api 1.0.0 sha256:10000c57bd3adc45a796199743b8b52e858b54d996ca34433f208c6b7b778e4c 436f25f95b39 18 minutes ago 179MB
bc653d069cc0955cb23470b3a82c197409ed69e4783eb8cf779f8349c34de5f7
{"status":"healthy","application":"student-ml-api","application_version":"1.1.0","model_version":"model-1"}
student-ml-api
784860d7d340a1a6851052a82f37a5ba75390f7ddd4adf9798340e8f6eb89585
{"status":"healthy","application":"student-ml-api","version":"1.0.0"}
mahad1090@mahad-1090:~/Desktop/student-ml-api$
```

Fig 21 — docker images --digests : latest and 1.1.0 share digest 2c9ccbd1... ; 1.0.0 keeps digest 10000c57... and remains available.

15. Rollback Exercise (Part 20)

Assume 1.1.0 has a production issue. Restore 1.0.0 with no source change and no rebuild:

```
docker rm -f student-ml-api
docker run -d --name student-ml-api -p 5005:5000 ghcr.io/mahad1090/student-ml-api:1.0.0
curl localhost:5005/health → {"status":"healthy","application":"student-ml-api","version":"1.0.0"}
```

Why this beats `git clone → pip install → python app.py` : the `1.0.0` image is the *exact bytes* that were tested and previously ran — same OS packages, same pinned Python, same wheels. A fresh `pip install` can resolve different transitive versions and reintroduce bugs. Rollback is one `docker run` (seconds, no build toolchain on the host) and is auditable by digest.

16. Traceability Challenge (Part 21)

Version 1.0.0

```
Pull Request : #1    feat: prediction API with CI pipeline
Merge Commit : fa0394b678fb878b4ec9d3ffb4454b26c00ee91d
Git Tag      : v1.0.0
Docker Image : ghcr.io/mahad1090/student-ml-api:1.0.0    (also :fa0394b)
Image Digest : sha256:10000c57bd3adc45a796199743b8b52e858b54d996ca34433f208c6b7b778e4c
```

Version 1.1.0

```
Pull Request : #2    feat: report application and model version on /health
Merge Commit : 02dc3db8bf199a185b6f0c9d0c454fb599e25e12
Git Tag      : v1.1.0
Docker Image : ghcr.io/mahad1090/student-ml-api:1.1.0    (also :latest, :02dc3db)
Image Digest : sha256:2c9ccbd13d38ad9e8fbb4a067825f32b2c39b09082b31e4372f03332baadcaf5
```

The image additionally carries the commit as an OCI label (`org.opencontainers.image.revision`), so a running container can always be traced to its source commit (Fig 15).

```

mahad1090@mahad-1090:~/Desktop/student-ml-api$ cd ~/Desktop/student-ml-api
git checkout main
git pull
git log --oneline --decorate -8
git tag
git show v1.1.0 --stat | head -20
Already on 'main'
Your branch is up to date with 'origin/main'.
Already up to date.
4650a2b (HEAD -> main, origin/main, origin/HEAD) docs: MLOps workflow write-up, traceability and failure analysis (#3)
02dc3db (tag: v1.1.0) feat: report application and model version on /health (#2)
fa8394b (tag: v1.0.0) feat: prediction API with CI pipeline (#1)
a8072f5 Initial commit
v1.0.0
v1.1.0
commit 02dc3db8bf199a185b6f0c9d0c454fb599e25e12
Author: Mahad Malik <168192717+Mahad1090@users.noreply.github.com>
Date: Thu Sep 10 20:41:24 2026 +0500

    feat: report application and model version on /health (#2)

    * feat: expose application and model versions on /health

    * test: assert application_version and model_version in health

-----

Co-authored-by: Mahad Malik <mahadmalik1090@gmail.com>

VERSION          | 2 +-
app.py            | 7 +++++
tests/test_app.py | 5 +++-
3 files changed, 10 insertions(+), 4 deletions(-)
mahad1090@mahad-1090:~/Desktop/student-ml-api$

```

Fig 17 — `git log --decorate + git tag :tags on main` , PR-merge commits.

17. CI vs Release, Metadata, Commit Tag, Cache (Parts 22–25)

CI workflow vs Release workflow (Part 22)

	CI — <code>ci.yml</code>	Release — <code>release.yml</code>
Trigger	<code>pull_request</code> → <code>main</code> , feature-branch push	push tag <code>v*.*.*</code>
Does	test, validate, build-check	test, build, version, publish
Publishes artifact	No	Yes

Why not publish an image from every PR? A PR is unreviewed and possibly unstable (or from a fork); publishing pollutes the registry with throwaway tags and risks deploying an un-merged change. Registry storage/bandwidth cost money and most PR builds are never used. Publishing should be a deliberate, tagged, auditable event — and a push credential is far more dangerous in a PR-triggered job than in a tag-triggered one only maintainers can fire.

Image metadata (Part 23)

```

org.opencontainers.image.title    = student-ml-api
org.opencontainers.image.version  = 1.1.0
org.opencontainers.image.revision = 02dc3db8bf199a185b6f0c9d0c454fb599e25e12

```

```
org.opencontainers.image.source    = https://github.com/Mahad1090/student-ml-api
org.opencontainers.image.created  = <build timestamp>
```

Verified with `docker inspect --format '{{json .Config.Labels}}'` (Fig 15).

Commit-SHA tag (Part 24)

Alongside `1.1.0` and `latest`, the release publishes `student-ml-api:02dc3db`. **Benefit:** a semantic tag is a human label that could be re-pushed and `latest` moves constantly; a commit-SHA tag is 1:1 with an immutable point in source history — given a running container you can name the exact commit, and given a commit you can pull exactly that image.

Build cache (Part 25)

With `COPY requirements.txt` → `RUN pip install` → `COPY app.py`: editing **only** `app.py` reuses the cached dependency layer (---> Using cache) and rebuilds in seconds; editing `requirements.txt` invalidates that layer and *everything after it*, so `pip install` re-runs. `COPY . .` before `pip install` is worse because any one-character source change forces a full re-install on every build — wasting CI minutes on every commit.

18. Failure Analysis (Part 26)

Failure 1 — Failed `pytest` blocks the PR

Symptom	PR #1 CI red; Unit tests failed, Docker build validation skipped; merge blocked.
Root cause	Assertion changed to <code>== "wrong"</code> while the endpoint returns <code>"healthy"</code> .
Evidence	Run 34494638490: <code>AssertionError: assert 'healthy' == 'wrong'</code> (Fig 5).
Correction	Restored the assertion; commit 3f9d86f ; run 34494921761 green (Fig 6).

Failure 2 — Missing dependency (deliberately reproduced)

Symptom	On branch <code>chore/failure-demo</code> , CI Unit tests failed during collection; Docker build validation skipped.
Root cause	<code>httpx</code> removed from <code>requirements.txt</code> . FastAPI's <code>TestClient</code> wraps <code>httpx</code> , so every test fails to import even though <code>app.py</code> still runs.
Evidence	Run 34498295688: <code>ModuleNotFoundError: No module named 'httpx'</code> / <code>RuntimeError: The starlette.testclient module requires the httpx package</code> (Fig 25).
Correction	Restored <code>httpx==0.28.1</code> ; branch discarded (never merged). A runtime-only missing dep (e.g. dropping <code>uvicorn</code>) would instead pass <code>pytest</code> but fail the container's <code>CMD</code> at start — which is why CI also builds the image.

Failure 3 — Wrong / already-allocated container port (observed locally)

Symptom	<code>docker run -p 5000:5000</code> → Bind for 0.0.0.0:5000 failed: port is already allocated . <code>curl localhost:5000/health</code> returned a <i>different</i> app's JSON.
Root cause	Three containers from an unrelated exercise already published host ports 5000–5002; the new container could not bind and the port that answered belonged to another service.
Evidence	<code>docker ps</code> showed 0.0.0.0:5000->5000/tcp on <code>competent_rhodes</code> , ...5001 / ...5002 on <code>staging-a/-b</code> ; our container stuck in <i>Created</i> .
Correction	Published on a free host port: <code>-p 5005:5000</code> . <code>curl localhost:5005/health</code> then returned the correct <code>student-ml-api</code> payload. The <i>container</i> port (<code>EXPOSE 5000</code> , <code>--port 5000</code>) is fixed by the image; only the <i>host</i> side of <code>-p host:container</code> collided.

Related — binding to 127.0.0.1 : if the app bound 127.0.0.1 instead of 0.0.0.0 , it would only accept connections from inside the container's namespace, so `-p 5005:5000` forwarding from the host would get *connection reset*. This is why the `CMD` uses `--host 0.0.0.0` .

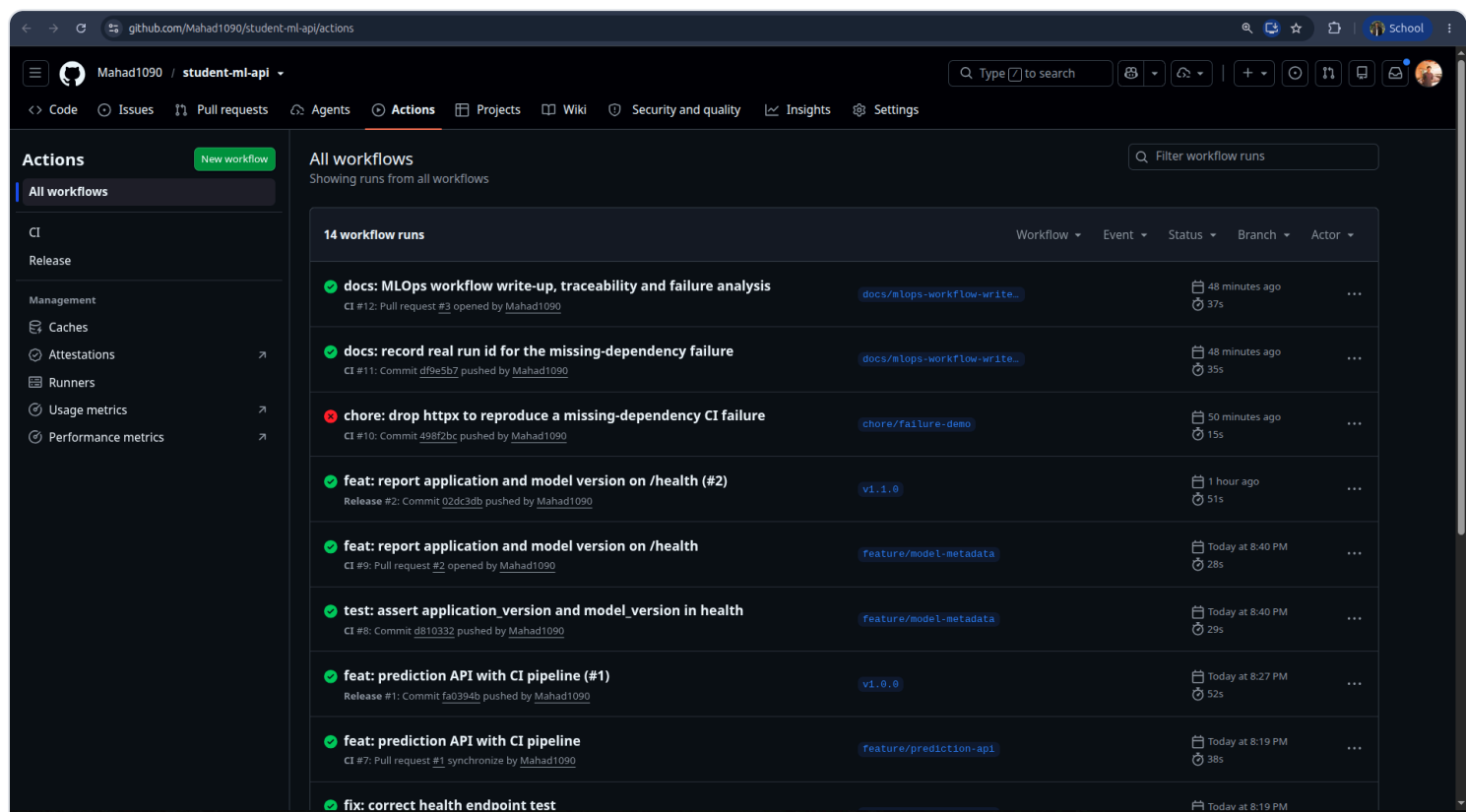


Fig 25 — Actions history: the **failed** chore/failure-demo and Part-6 runs among the green CI / Release runs.

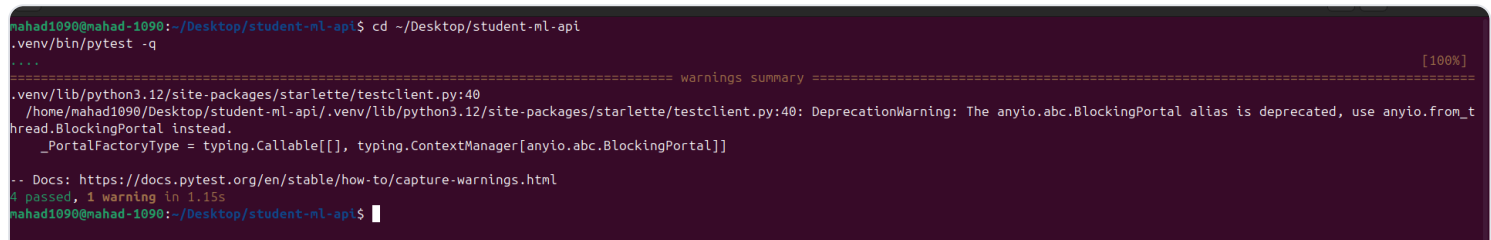
19. Viva Questions

1. **Why avoid pushing directly to main ?** No review, no CI gate, no audit trail — a bad commit is instantly "production". PRs make every change reviewed, tested and revertible as a unit.

2. **Purpose of a PR beyond merging?** Code review, discussion, a CI checkpoint, a documented rationale, and a durable link between a change and why it was made.
3. **Why CI before merge?** To keep `main` always green. A failure caught after merge means everyone pulls broken code and the fix needs its own PR.
4. **Image vs container?** An image is an immutable, layered filesystem + metadata (a build artifact). A container is a running/stopped instance of an image with its own writable layer, PID, network and lifecycle.
5. **Why version images?** So a deploy names an exact artifact, rollbacks are possible, and "what is running?" has a precise answer.
6. **Why is `latest` insufficient?** It is a moving pointer — different bytes over time, recorded nowhere; two machines pulling it a day apart can run different code.
7. **Why promote the same artifact instead of rebuilding?** A rebuild can pull different transitive dependencies / base-image patches and produce an untested image. Promotion ships the tested bits.
8. **Purpose of a container registry?** A versioned, access-controlled store that distributes immutable image artifacts (by tag and by digest) to any environment.
9. **CI vs release workflow?** CI validates proposed changes on PRs and publishes nothing. Release runs on a version tag and is the only thing that builds and pushes a published image.
10. **Why secrets for registry credentials?** Hard-coded credentials live in git history forever, are visible to anyone with read access, and cannot be rotated cleanly. Secrets are injected at runtime, masked in logs, and scoped.
11. **How to identify the commit that produced an image?** The `<short-sha>` image tag, or `docker inspect --format '{{.Config.Labels.org.opencontainers.image.revision}}'` label.
12. **Why does layer ordering affect CI/CD performance?** A changed layer invalidates the cache for itself and everything after it — put slow, rarely-changing steps (dependency install) before fast, frequently-changing ones (app code).
13. **How to roll back 1.1.0 → 1.0.0?** `docker run ghcr.io/mahad1090/student-ml-api:1.0.0` — the old image is still in the registry; no code change, no rebuild.
14. **Git tag vs Docker image tag?** The Git tag (`v1.1.0`) marks a commit in source history and triggers the release; the Docker tag (`1.1.0`) labels the resulting image in the registry. The workflow derives the second from the first.
15. **App version vs model version changing independently?** Both must be recorded per release (here via `application_version` + `model_version`) — the same code with a different model, or the same model behind different code, are different systems. It also multiplies the test matrix and complicates rollback (roll back code, model, or both?).

20. Screenshot Appendix

All evidence images, in order. (Figures are also referenced inline above.)



```
mahad1090@mahad-1090:~/Desktop/student-ml-api$ cd ~/Desktop/student-ml-api
.venv/bin/pytest -q
.....
===== warnings summary ===== [100%]
.venv/lib/python3.12/site-packages/starlette/testclient.py:40
/home/mahad1090/Desktop/student-ml-api/.venv/lib/python3.12/site-packages/starlette/testclient.py:40: DeprecationWarning: The anyio.abc.BlockingPortal alias is deprecated, use anyio.from_thread.BlockingPortal instead.
  _PortalFactoryType = typing.Callable[[], typing.ContextManager[anyio.abc.BlockingPortal]]

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
4 passed, 1 warning in 1.15s
mahad1090@mahad-1090:~/Desktop/student-ml-api$
```

Fig 1 — `pytest -q` : 4 passed.

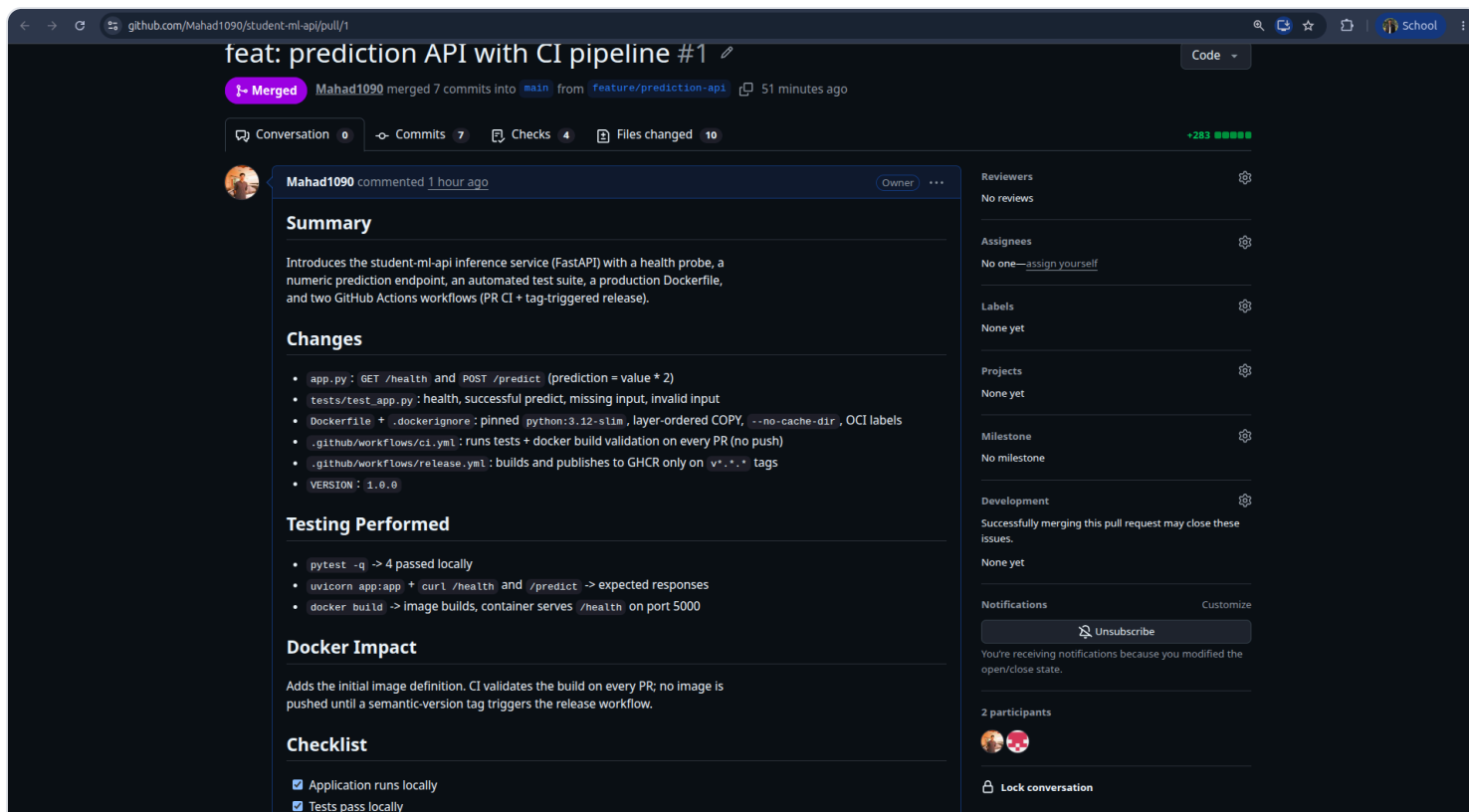


Fig 2 — PR #1 description & checklist.

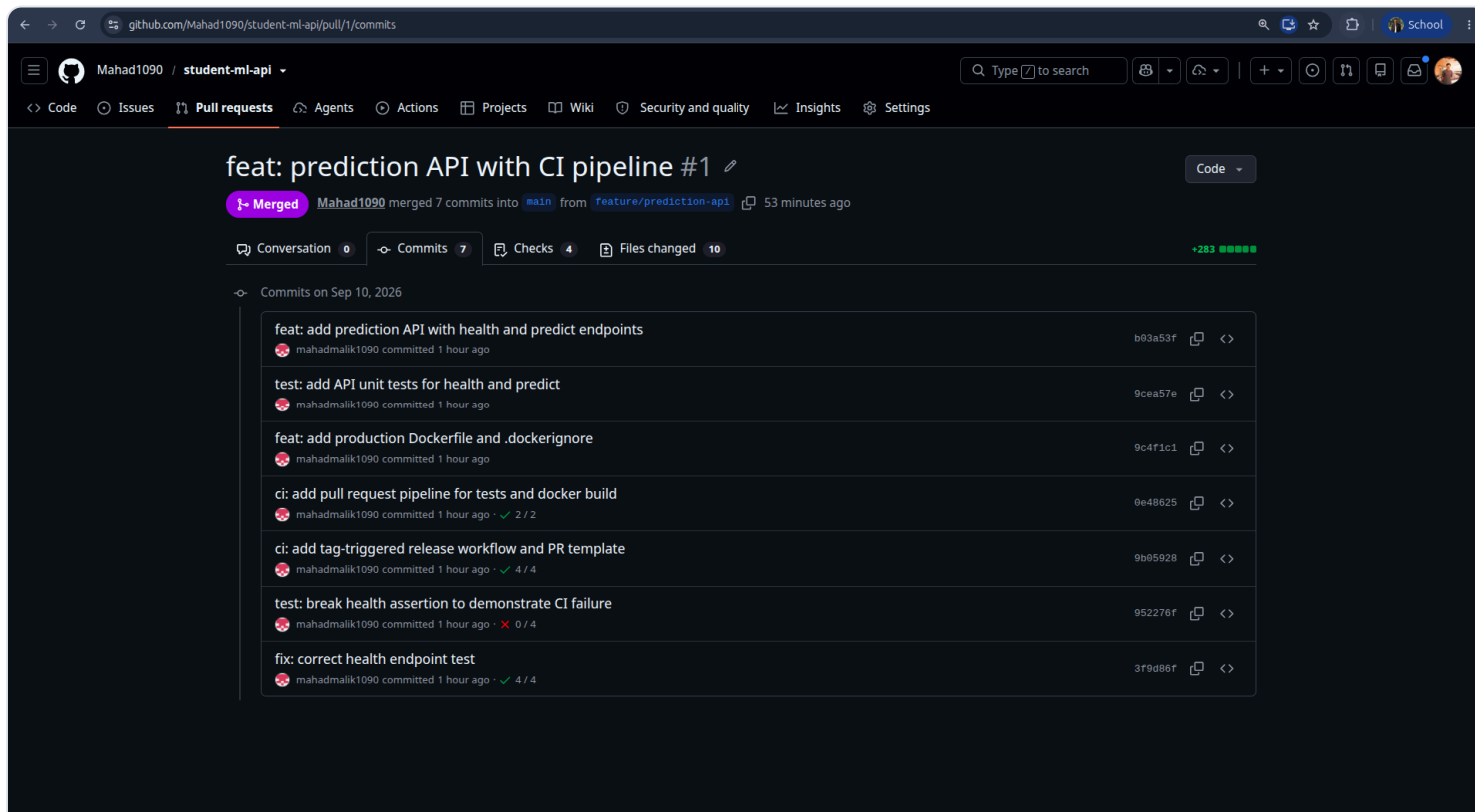


Fig 3 — PR #1 commits (break ✗ / fix ✓).

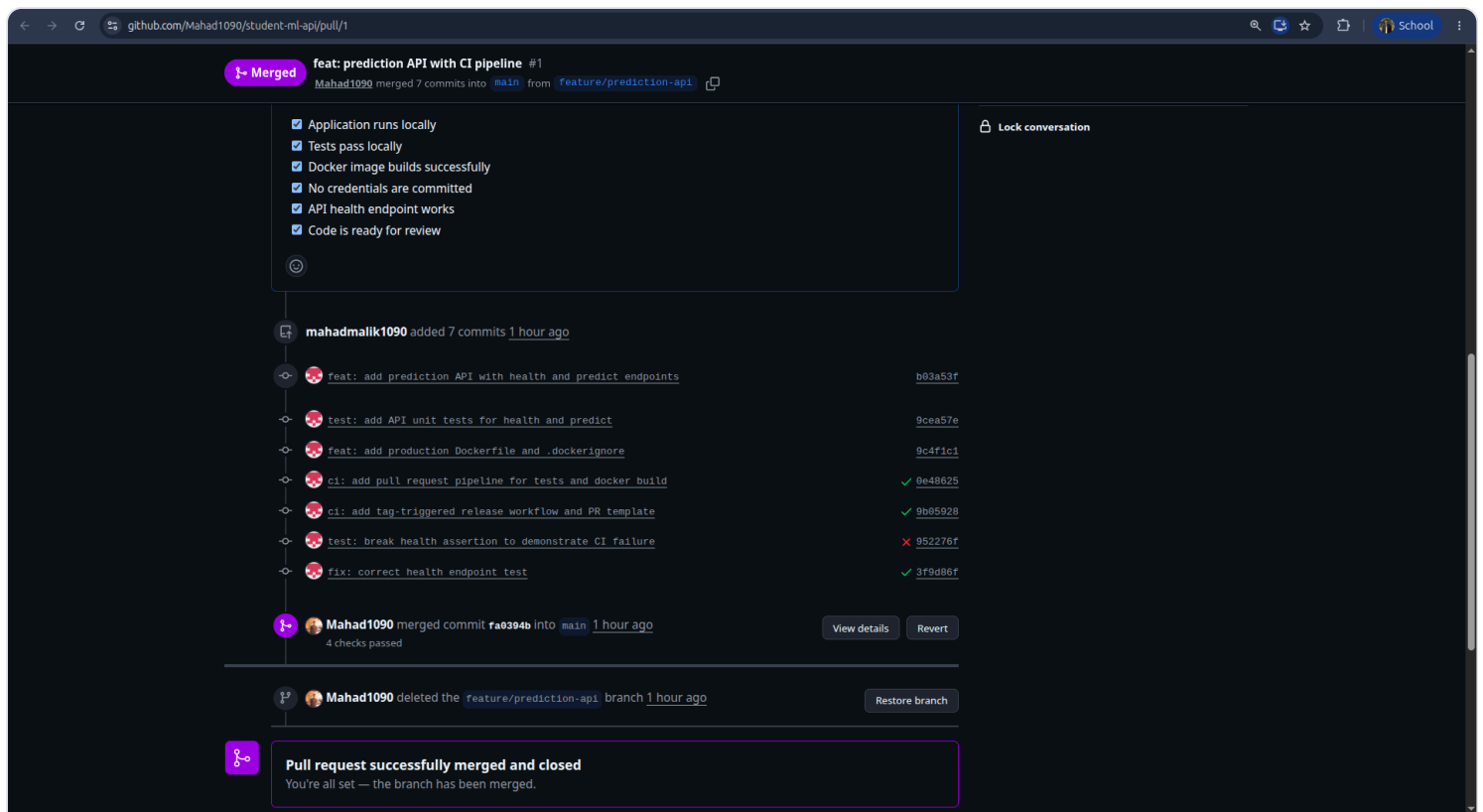


Fig 4 — PR #1 squash-merged (fa0394b).

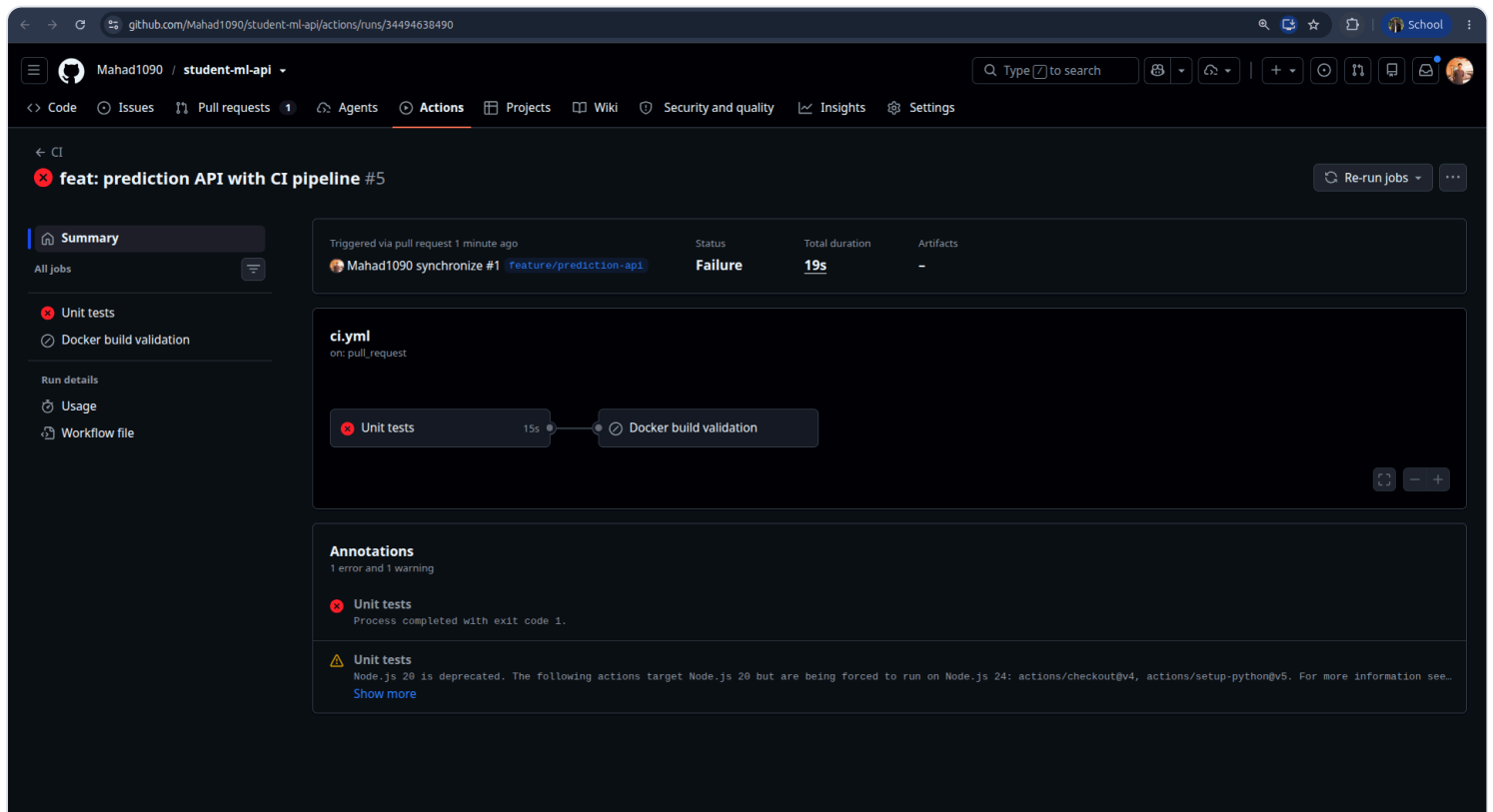


Fig 5 — Part 6: CI FAILED.

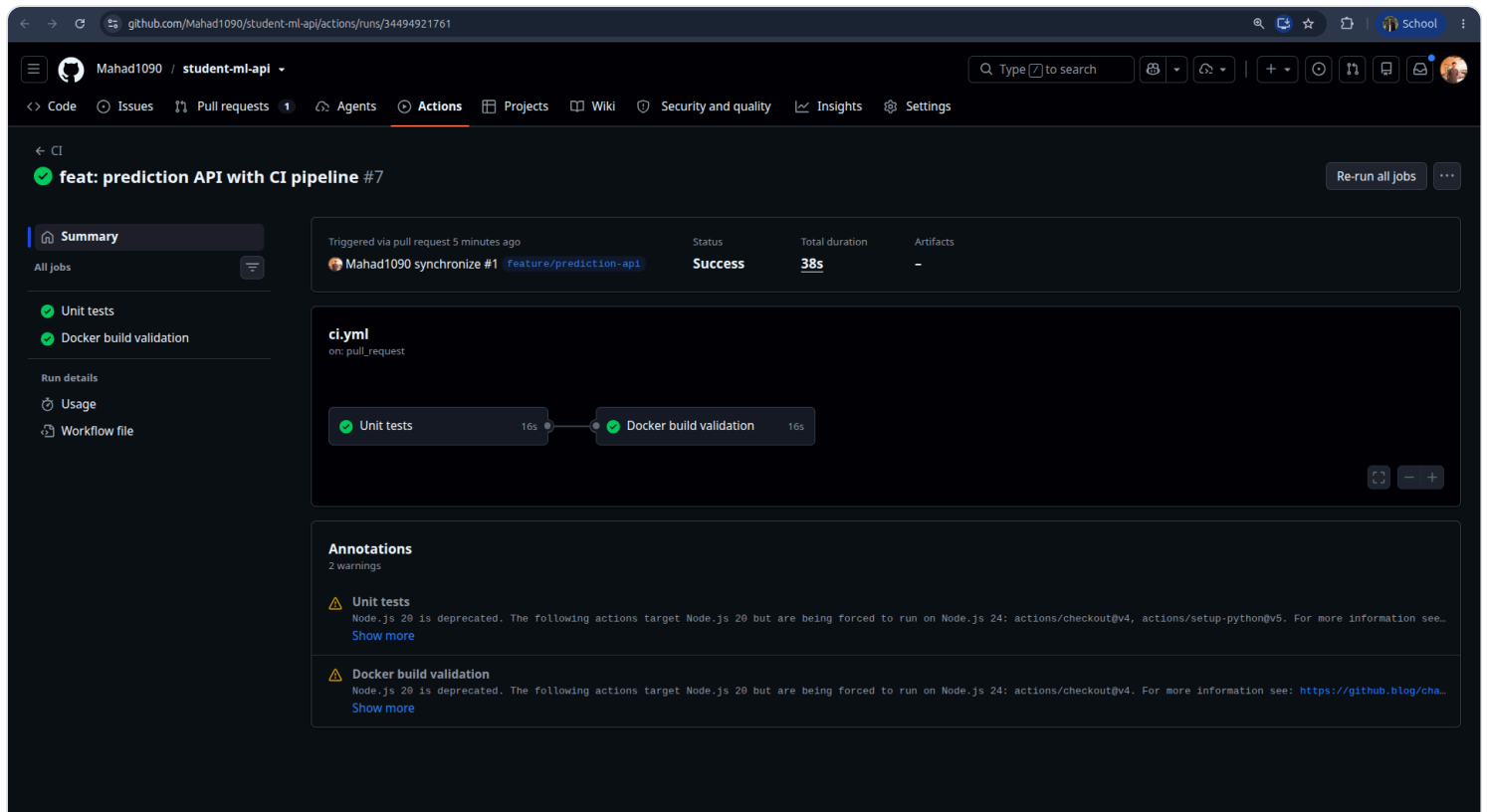


Fig 6 — Part 6: CI SUCCESS after fix.

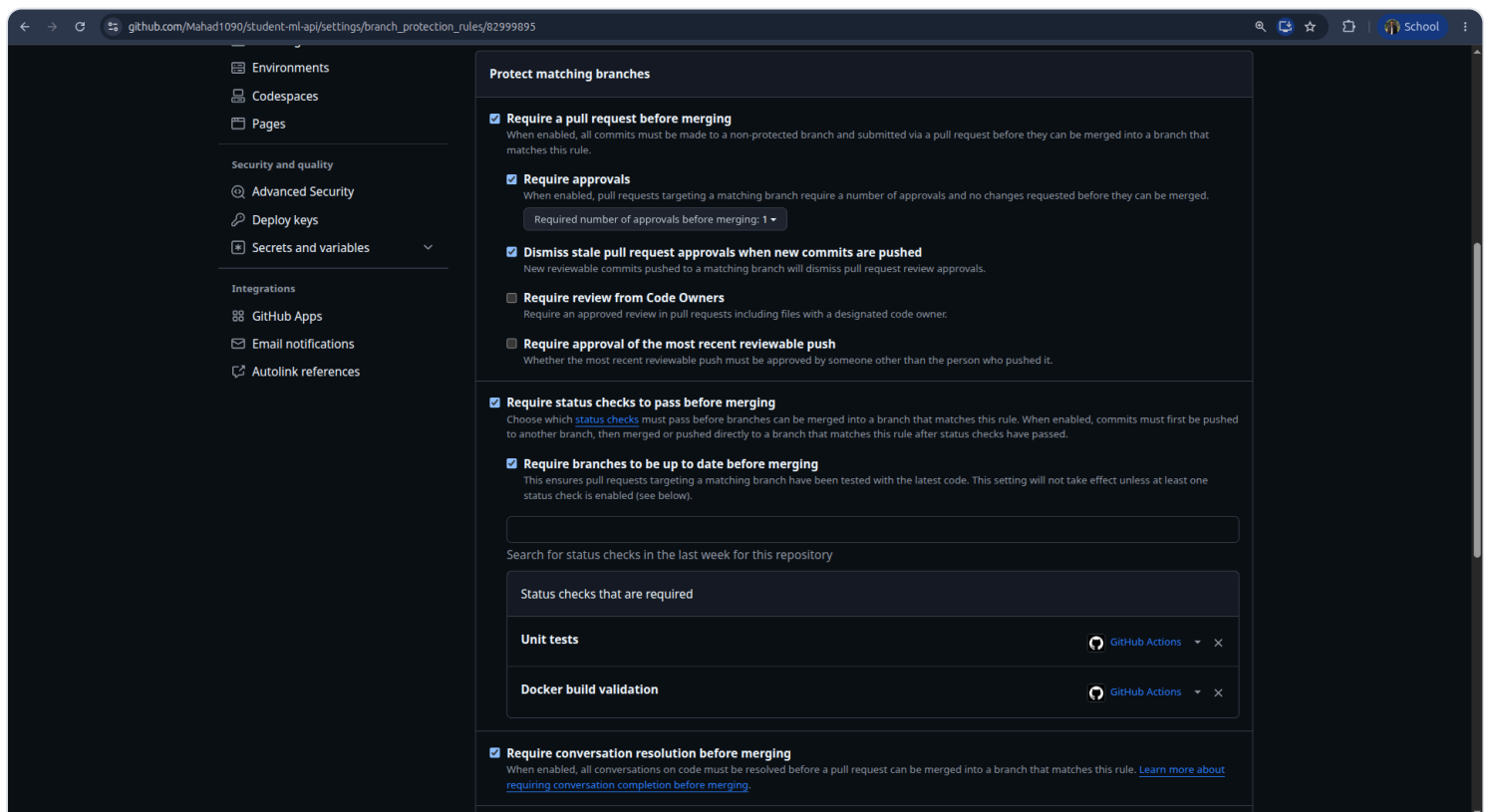


Fig 7 — main branch protection.

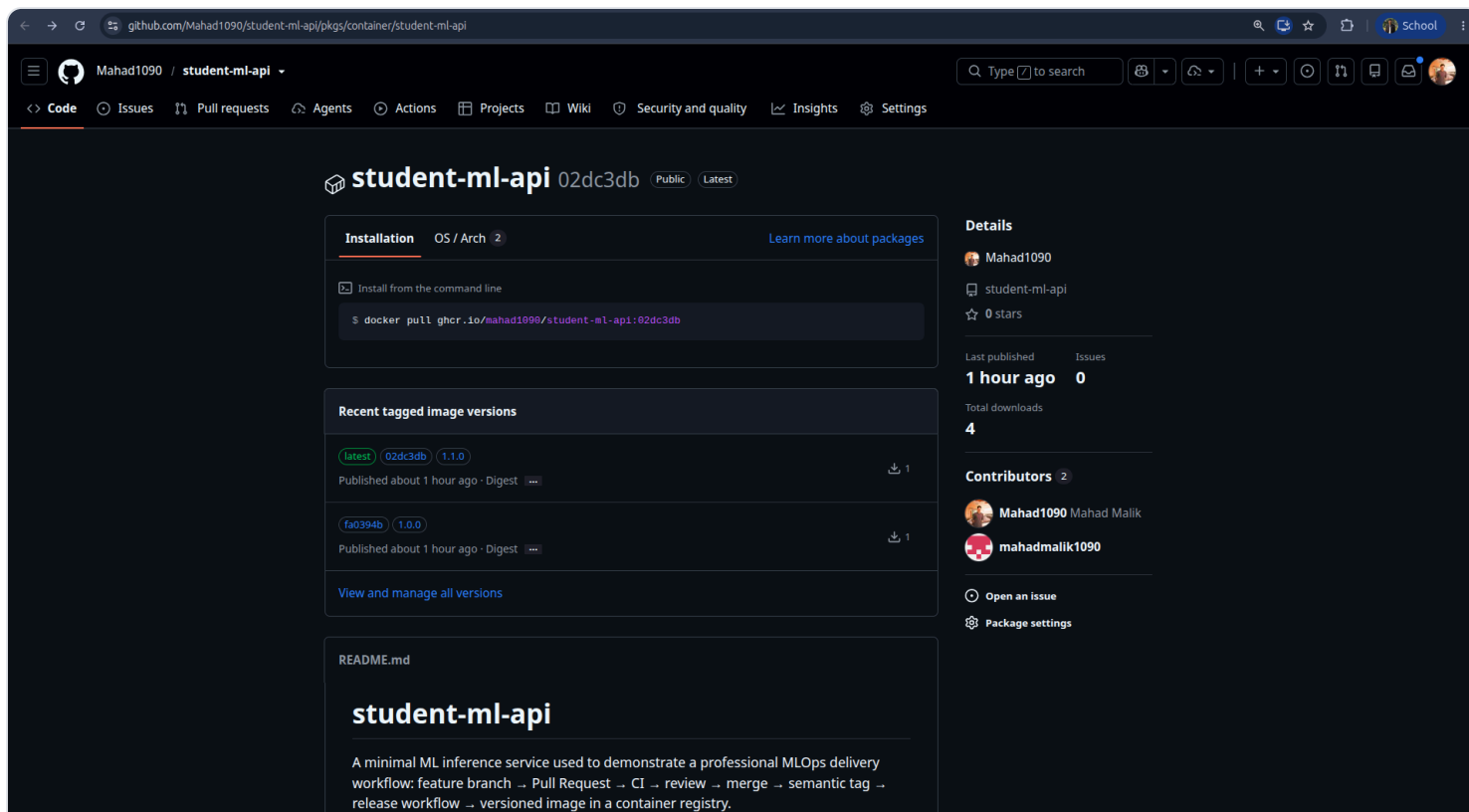


Fig 8 — GHCR package tags.

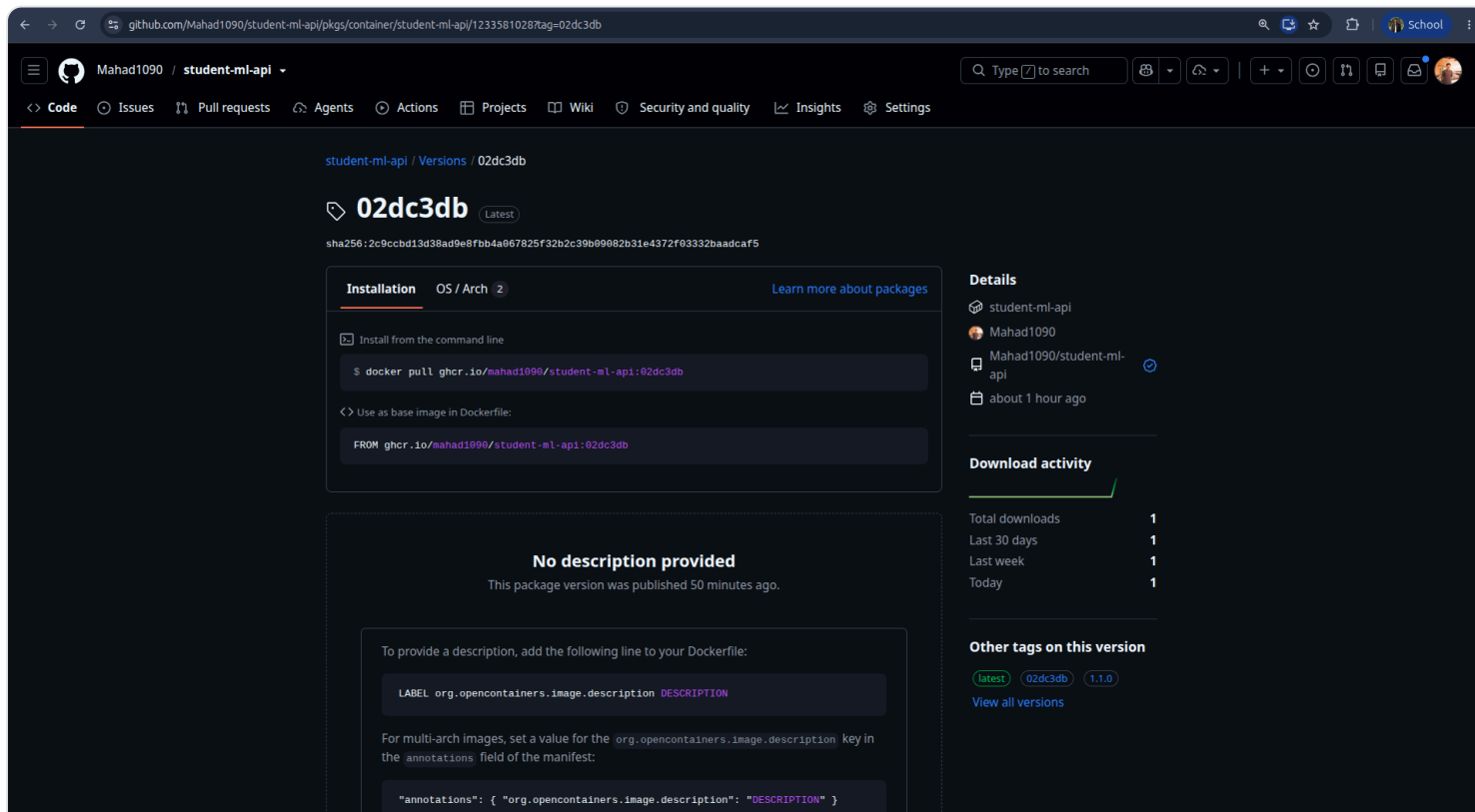
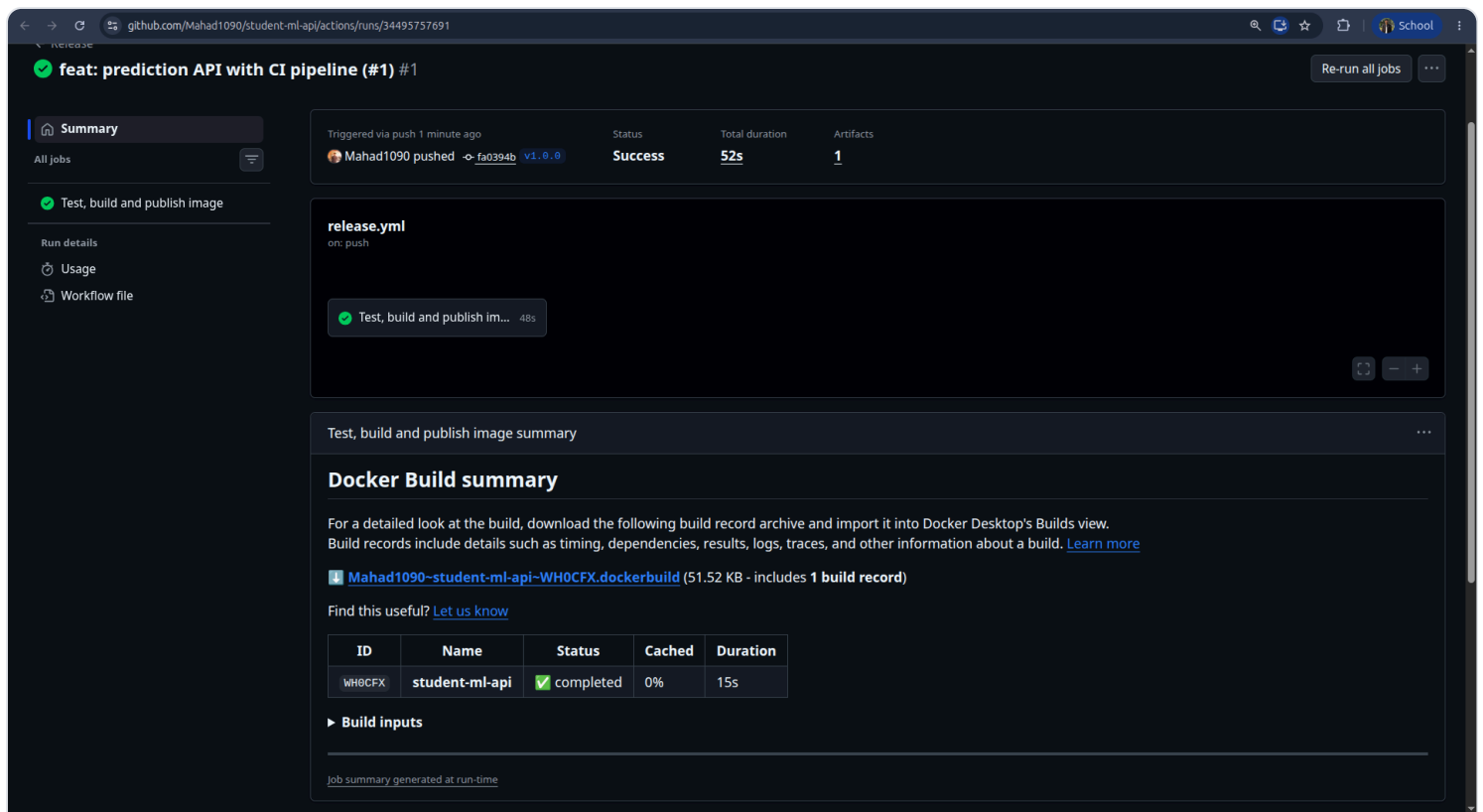


Fig 9 — image digest for 1.1.0 / latest .



The screenshot shows the GitHub Actions interface for a workflow named "feat: prediction API with CI pipeline (#1) #1". The workflow is triggered by a push to the "v1.0.0" branch. The status is "Success" with a total duration of 52s and 1 artifact. The workflow file is "release.yml". The job "Test, build and publish image" is shown with a summary section titled "Test, build and publish image summary". Below this is a "Docker Build summary" section, which includes a link to the build record archive and a table of build records.

Triggered via push 1 minute ago

Mahad1090 pushed · fa0394b · v1.0.0

Status: Success

Total duration: 52s

Artifacts: 1

release.yml

on: push

Test, build and publish image summary

Docker Build summary

For a detailed look at the build, download the following build record archive and import it into Docker Desktop's Builds view. Build records include details such as timing, dependencies, results, logs, traces, and other information about a build. [Learn more](#)

📄 Mahad1090-student-ml-api-WH0CFX.dockerbuild (51.52 KB - includes 1 build record)

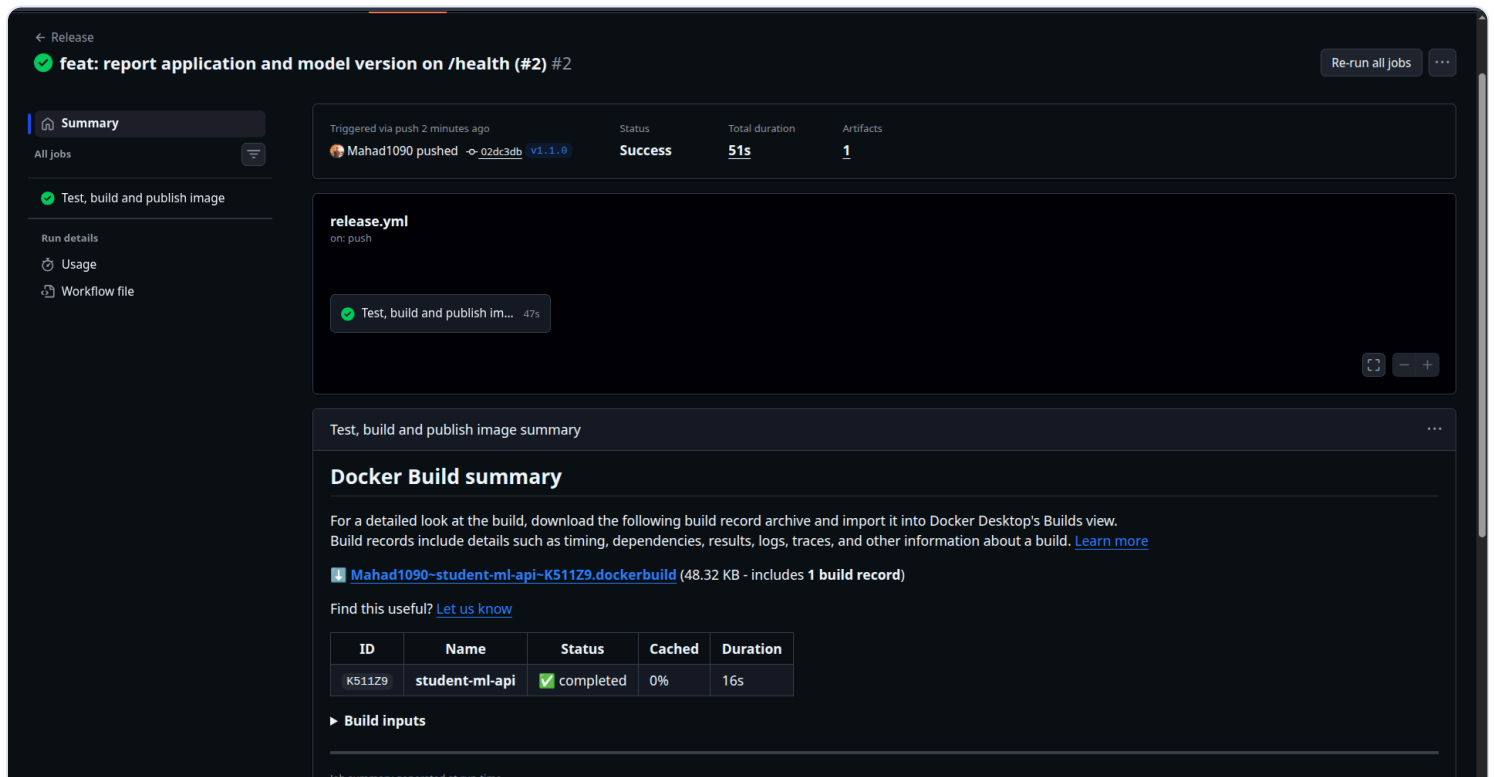
Find this useful? [Let us know](#)

ID	Name	Status	Cached	Duration
WH0CFX	student-ml-api	✅ completed	0%	15s

► Build inputs

job summary generated at run-time

Fig 10 — Release workflow for v1.0.0 .



The screenshot shows the GitHub Actions interface for a workflow named "feat: report application and model version on /health (#2) #2". The workflow is triggered by a push to the "v1.1.0" branch. The status is "Success" with a total duration of 51s and 1 artifact. The workflow file is "release.yml". The job "Test, build and publish image" is shown with a summary section titled "Test, build and publish image summary". Below this is a "Docker Build summary" section, which includes a link to the build record archive and a table of build records.

← Release

feat: report application and model version on /health (#2) #2

Triggered via push 2 minutes ago

Mahad1090 pushed · 02dc3db · v1.1.0

Status: Success

Total duration: 51s

Artifacts: 1

release.yml

on: push

Test, build and publish image summary

Docker Build summary

For a detailed look at the build, download the following build record archive and import it into Docker Desktop's Builds view. Build records include details such as timing, dependencies, results, logs, traces, and other information about a build. [Learn more](#)

📄 Mahad1090-student-ml-api-K51129.dockerbuild (48.32 KB - includes 1 build record)

Find this useful? [Let us know](#)

ID	Name	Status	Cached	Duration
K51129	student-ml-api	✅ completed	0%	16s

► Build inputs

job summary generated at run-time

Fig 11 — Release workflow for v1.1.0 .

```

mahad1090@mahad-1090:~/Desktop/student-ml-api$ cd ~/Desktop/student-ml-api
docker build -t student-ml-api:1.0.0 \
  --build-arg APP_VERSION=1.0.0 \
  --build-arg GIT_COMMIT=$(git rev-parse HEAD) \
  --build-arg BUILD_DATE=$(date -u +%Y-%m-%dT%H:%M:%SZ) .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 14.34kB
Step 1/11 : FROM python:3.12-slim
--> ec7d6c95cd36
Step 2/11 : ARG APP_VERSION=dev
--> Running in 0a5ca096f6ca
--> Removed intermediate container 0a5ca096f6ca
--> 9e8513aca92a
Step 3/11 : ARG GIT_COMMIT=unknown
--> Running in bc0821fe77e8
--> Removed intermediate container bc0821fe77e8
--> d5cc2dd6dca0
Step 4/11 : ARG BUILD_DATE=unknown
--> Running in 0591428e4bc1
--> Removed intermediate container 0591428e4bc1
--> 71a78aa952a9
Step 5/11 : LABEL org.opencontainers.image.title="student-ml-api"      org.opencontainers.image.description="Minimal ML inference service"      org.opencontainers.image.version="${APP_VERSION}"
             org.opencontainers.image.revision="${GIT_COMMIT}"        org.opencontainers.image.source="https://github.com/Mahad1090/student-ml-api"      org.opencontainers.image.created="${BUILD_DATE}"
--> Running in 2929fed6f381
--> Removed intermediate container 2929fed6f381
--> 2f2d17e879d8
Step 6/11 : WORKDIR /app
--> Running in ac89adb75cf9
--> Removed intermediate container ac89adb75cf9
--> 65a62166ce4b
Step 7/11 : COPY requirements.txt .
--> 212670e61192
Step 8/11 : RUN pip install --no-cache-dir -r requirements.txt
--> Running in 88361ad0780d
Collecting fastapi==0.115.6 (from -r requirements.txt (line 1))
  Downloading fastapi-0.115.6-py3-none-any.whl.metadata (27 kB)
Collecting uvicorn==0.34.0 (from uvicorn[standard]==0.34.0->-r requirements.txt (line 2))
  Downloading uvicorn-0.34.0-py3-none-any.whl.metadata (6.5 kB)
Collecting pydantic==2.10.4 (from -r requirements.txt (line 3))
  Downloading pydantic-2.10.4-py3-none-any.whl.metadata (29 kB)
Collecting pytest==8.3.4 (from -r requirements.txt (line 4))
  Downloading pytest-8.3.4-py3-none-any.whl.metadata (7.5 kB)
Collecting httpx==0.28.1 (from -r requirements.txt (line 5))
  Downloading httpx-0.28.1-py3-none-any.whl.metadata (7.4 kB)

```

Fig 12 — docker build (steps 1–8).

```

Downloading httpcore-1.0.9-py3-none-any.whl (78 kB)
Downloading pydantic_core-2.27.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.0 MB)
 2.0/2.0 MB 1.4 MB/s eta 0:00:00
Downloading annotated_types-0.8.0-py3-none-any.whl (13 kB)
Downloading click-8.5.0-py3-none-any.whl (125 kB)
Downloading h11-0.16.0-py3-none-any.whl (37 kB)
Downloading httptools-0.8.0-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (523 kB)
Downloading pluggy-1.6.0-py3-none-any.whl (20 kB)
Downloading python_dotenv-1.2.3-py3-none-any.whl (22 kB)
Downloading pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (807 kB)
 807.9/807.9 kB 1.6 MB/s eta 0:00:00
Downloading starlette-0.41.3-py3-none-any.whl (73 kB)
Downloading anyio-4.15.1-py3-none-any.whl (132 kB)
Downloading idna-3.19-py3-none-any.whl (68 kB)
Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
Downloading uvloop-0.22.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (4.4 MB)
 4.4/4.4 MB 1.7 MB/s eta 0:00:00
Downloading watchfiles-1.2.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (456 kB)
Downloading websockets-17.1-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (224 kB)
Downloading certifi-2026.7.22-py3-none-any.whl (136 kB)
Downloading iniconfig-2.3.0-py3-none-any.whl (7.5 kB)
Downloading packaging-26.3-py3-none-any.whl (129 kB)
Installing collected packages: websockets, uvloop, typing-extensions, pyyaml, python-dotenv, pluggy, packaging, iniconfig, idna, httptools, h11, click, certifi, annotated-types, uvicorn, pytest, pydantic-core, httpcore, anyio, watchfiles, starlette, pydantic, httpx, fastapi
Successfully installed annotated-types-0.8.0 anyio-4.15.1 certifi-2026.7.22 click-8.5.0 fastapi-0.115.6 h11-0.16.0 httpcore-1.0.9 httptools-0.8.0 httpx-0.28.1 idna-3.19 iniconfig-2.3.0 packaging-26.3 pluggy-1.6.0 pydantic-2.10.4 pydantic-core-2.27.2 pytest-8.3.4 python-dotenv-1.2.3 pyyaml-6.0.3 starlette-0.41.3 typing-extensions-4.16.0 uvicorn-0.34.0 uvloop-0.22.1 watchfiles-1.2.0 websockets-17.1
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager, possibly rendering your system unusable. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv. Use the --root-user-action option if you know what you are doing and want to suppress this warning.

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
--> Removed intermediate container 88361ad0780d
--> 9abf6e77cb7a
Step 9/11 : COPY app.py VERSION ./
--> 76fa7e7ed68a
Step 10/11 : EXPOSE 5000
--> Running in bd6fcc62c938
--> Removed intermediate container bd6fcc62c938
--> 11575edf4294
Step 11/11 : CMD ["uvicorn", "app:app", "--host", "0.0.0.0", "--port", "5000"]
--> Running in a7b60a373707
--> Removed intermediate container a7b60a373707
--> daa754b53267
Successfully built daa754b53267
Successfully tagged student-ml-api:1.0.0
mahad1090@mahad-1090:~/Desktop/student-ml-api$

```

Fig 13 — docker build (steps 9–11, tagged).

```

mahad1090@mahad-1090:~/Desktop/student-ml-api$ docker rm -f student-ml-api
docker run -d --name student-ml-api -p 5005:5000 student-ml-api:1.0.0
sleep 1
curl http://localhost:5005/health
echo
curl -X POST localhost:5005/predict -H 'content-type: application/json' -d '{"value":10}'
student-ml-api
cd0c8599377dde5698d8e68768b37811f7dee9144398f0f8e38b2ce084d9561f
{"status":"healthy","application":"student-ml-api","version":"1.0.0"}
{"input":10.0,"prediction":20.0}mahad1090@mahad-1090:~/Desktop/student-ml-api$

```

Fig 14 — container from local image, endpoints verified.

```

mahad1090@mahad-1090:~/Desktop/student-ml-api$ docker images | grep -E 'student-ml-api|REPOSITORY'
docker ps --filter name=student-ml-api
docker logs student-ml-api
docker inspect student-ml-api --format 'Cmd={{.Config.Cmd}} WorkingDir={{.Config.WorkingDir}} Exposed={{.Config.ExposedPorts}}'
docker inspect student-ml-api --format 'Image={{.Image}} ContainerID={{.Id}}'
docker exec student-ml-api sh -c "pwd && ls -la && cat VERSION"
docker inspect student-ml-api --format '{{json .Config.Labels}}' | python3 -m json.tool
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
student-ml-api:1.0.0      b9bab104a906      179MB      0B      U
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAMES
cd0c8599377d   student-ml-api:1.0.0  "uvicorn app:app --h..."  54 seconds ago Up 54 seconds  0.0.0.0:5005->5000/tcp, [::]:5005->5000/tcp  student-ml-api
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:5000 (Press CTRL+C to quit)
INFO: 172.17.0.1:48594 - "GET /health HTTP/1.1" 200 OK
INFO: 172.17.0.1:48602 - "POST /predict HTTP/1.1" 200 OK
Cmd=[uvicorn app:app --host 0.0.0.0 --port 5000] WorkingDir=/app Exposed=map[5000/tcp:{}]
Image=sha256:b9bab104a90690d3ad6432eb86a0407215784bbf364573ce7aeed49405b5b5b7 ContainerID=cd0c8599377dde5698d8e68768b37811f7dee9144398f0f8e38b2ce084d9561f
/app
total 24
drwxr-xr-x 1 root root 4096 Sep 10 15:33 .
drwxr-xr-x 1 root root 4096 Sep 10 15:32 ..
-rw-rw-r-- 1 root root 6 Sep 10 15:27 VERSION
drwxr-xr-x 2 root root 4096 Sep 10 15:33 __pycache__
-rw-rw-r-- 1 root root 1623 Sep 10 15:27 app.py
-rw-rw-r-- 1 root root 88 Sep 10 15:27 requirements.txt
1.0.0
{
  "org.opencontainers.image.created": "2026-09-10T15:31:05Z",
  "org.opencontainers.image.description": "Minimal ML inference service",
  "org.opencontainers.image.revision": "fa0394b678fb878b4ec9d3fffb4454b26c00ee91d",
  "org.opencontainers.image.source": "https://github.com/Mahad1090/student-ml-api",
  "org.opencontainers.image.title": "student-ml-api",
  "org.opencontainers.image.version": "1.0.0"
}
mahad1090@mahad-1090:~/Desktop/student-ml-api$

```

Fig 15 — Part 11 inspection + OCI labels.

```

docker rmi student-ml-api:1.0.0

# pull the exact artifact the release workflow published
docker pull ghcr.io/mahad1090/student-ml-api:1.0.0
docker inspect --format '{{index .RepoDigests 0}}' ghcr.io/mahad1090/student-ml-api:1.0.0

# run the downloaded image
docker run -d --name student-ml-api -p 5005:5000 ghcr.io/mahad1090/student-ml-api:1.0.0
sleep 1
curl http://localhost:5005/health

WARNING! Your credentials are stored unencrypted in '/home/mahad1090/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
student-ml-api
Untagged: student-ml-api:1.0.0
Deleted: sha256:b9bab104a90690d3ad6432eb86a0407215784bbf364573ce7aeed49405b5b5b7
Deleted: sha256:530db45a1bdb456f08ba212cd2d2d17da97769e7f0d145c923b1a61816ff550
Deleted: sha256:4196da2ca7825f0d090b9ad92b5eb1a056cacc42542ca45b452f2e8c034096e
Deleted: sha256:0b152e3f699b1099a5336ccdb0bf6e308de2c11cd048704e95f753d0cf5066bc
Deleted: sha256:42abdbcb694983f16f78bdcdb6863201c96329d815e8bceb746d65d726970064
Deleted: sha256:bc75e5f8e49cbdb0bcc25bbfcaf3ae0ef110f69f84014f6c3c74d45f1eb2a2b07
Deleted: sha256:9c5b1a872ee2928002d27c7e8b8ba6b29bc63fba484f3572629513592292d8b5
Deleted: sha256:83cc2ed46b357499ea61b37c4b117fd77afdd7501f17904f7b9181e4b6b22481
Deleted: sha256:be918425435e61488aeb31b4f3399054a83df9c9cbcd1ee824e26eb41cee7b9b3
Deleted: sha256:85531d3bbd5c5eaa9098621d385daf7b749c8632cbb43f01845709d280639f2
Deleted: sha256:e788c7a508d6355c67e2af0d179e78330fb87a66f3e170548772b3fe13ab3256
Deleted: sha256:7b76cc7e13ea993a29ef731203c2938473a08063ac0004af2406f0c65f440f5d
Deleted: sha256:4b5bfc8de5dbb70e03f895598dc8b1b7b496b28d35aa1c61de8c6e7811873be1
Deleted: sha256:c7fe16b76e71d089731b7921e70fed364156abd5c7cd74059db57df1f27c5306
1.0.0: Pulling from mahad1090/student-ml-api
6310eb16bf42: Already exists
dbd0b7849e6c: Already exists
1560cfed01dc: Already exists
56235e424563: Already exists
5f9c87856cb7: Pull complete
a34f03daca35: Pull complete
650a66b0edce: Pull complete
7fa4cda34c04: Pull complete
Digest: sha256:10000c57bd3adc45a796199743b8b52e858b54d996ca34433f208c6b7b778e4c
Status: Downloaded newer image for ghcr.io/mahad1090/student-ml-api:1.0.0
ghcr.io/mahad1090/student-ml-api:1.0.0
ghcr.io/mahad1090/student-ml-api@sha256:10000c57bd3adc45a796199743b8b52e858b54d996ca34433f208c6b7b778e4c
42ffa09066c5bbe077e9afd40415156b76b9d8091842f4d1789378d0c42476bc
{"status":"healthy","application":"student-ml-api","version":"1.0.0"}mahad1090@mahad-1090:~/Desktop/student-ml-api$

```

Fig 16 — Part 17: rmi → pull → run from GHCR.

```

mahad1090@mahad-1090:~/Desktop/student-ml-api$ cd ~/Desktop/student-ml-api
git checkout main
git pull
git log --oneline --decorate -8
git tag
git show v1.1.0 --stat | head -20
Already on 'main'
Your branch is up to date with 'origin/main'.
Already up to date.
4650a2b (HEAD -> main, origin/main, origin/HEAD) docs: MLOps workflow write-up, traceability and failure analysis (#3)
02dc3db (tag: v1.1.0) feat: report application and model version on /health (#2)
fa0394b (tag: v1.0.0) feat: prediction API with CI pipeline (#1)
a8072f5 Initial commit
v1.0.0
v1.1.0
commit 02dc3db8bf199a185b6f0c9d0c454fb599e25e12
Author: Mahad Malik <168192717@Mahad1090@users.noreply.github.com>
Date: Thu Sep 10 20:41:24 2026 +0500

    feat: report application and model version on /health (#2)

    * feat: expose application and model versions on /health

    * test: assert application_version and model_version in health

-----

Co-authored-by: Mahad Malik <mahadmalik1090@gmail.com>

VERSION          | 2 +-
app.py            | 7 +++++
tests/test_app.py | 5 +---
3 files changed, 10 insertions(+), 4 deletions(-)
mahad1090@mahad-1090:~/Desktop/student-ml-api$

```

Fig 17 — git log / git tag on main.

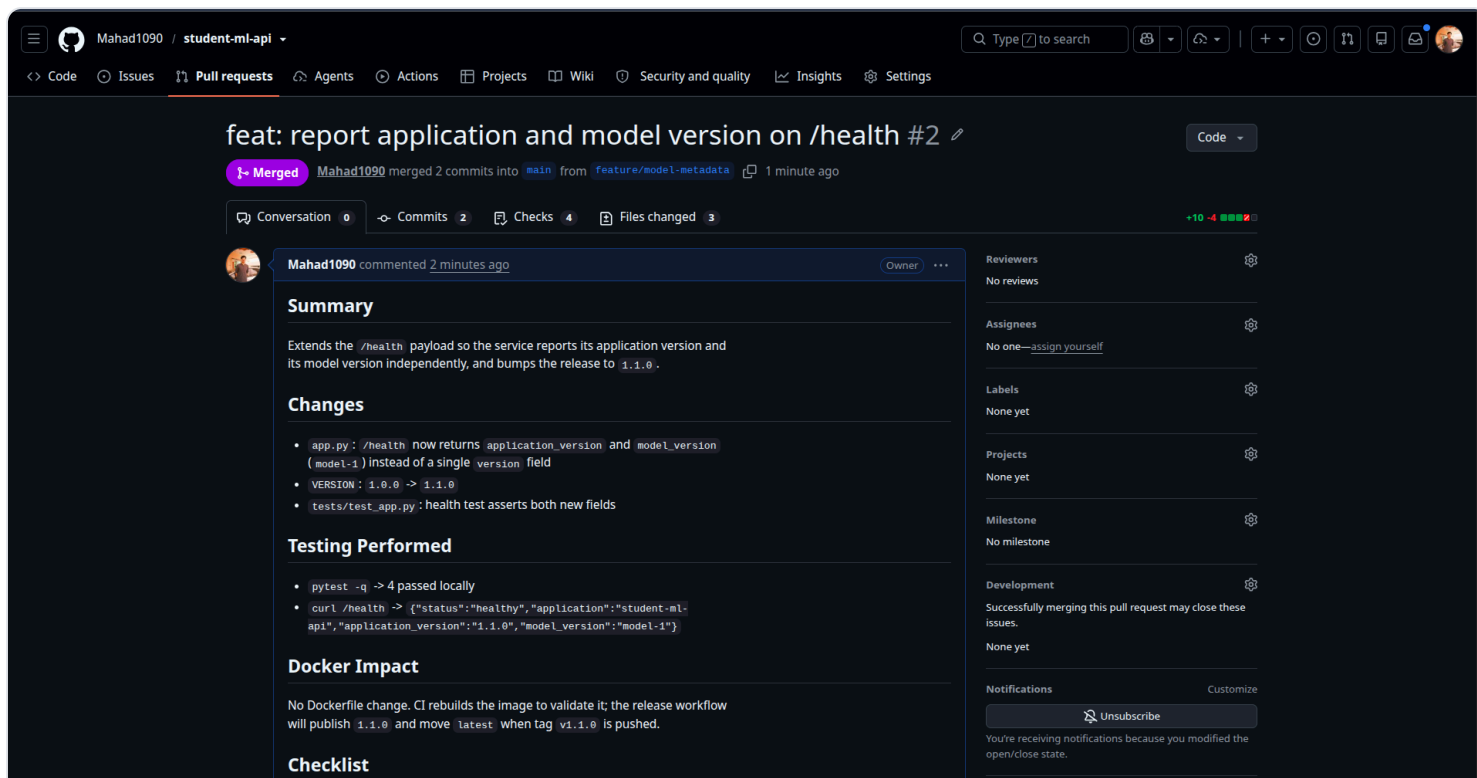


Fig 18 — PR #2 description.

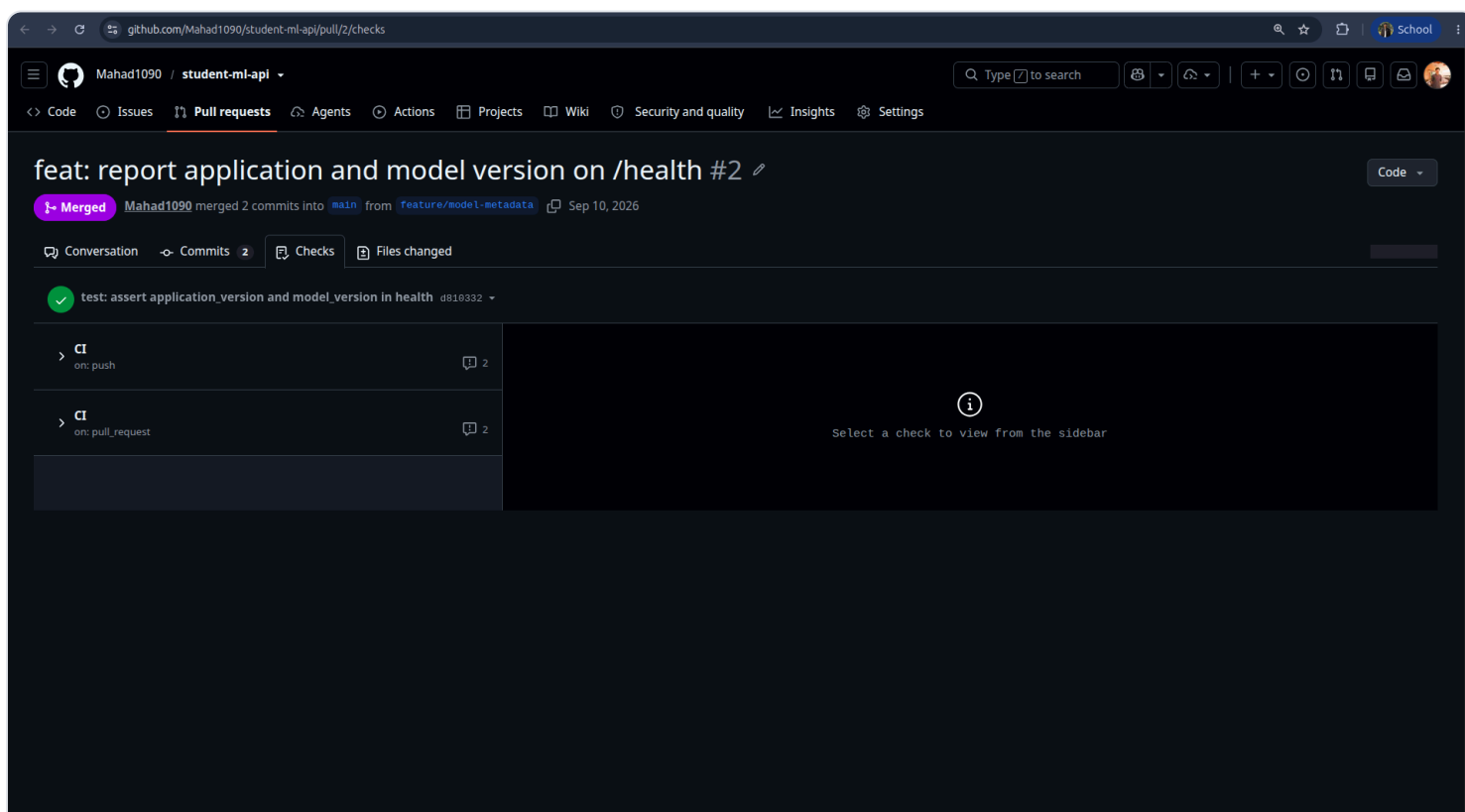


Fig 19 — PR #2 required checks.

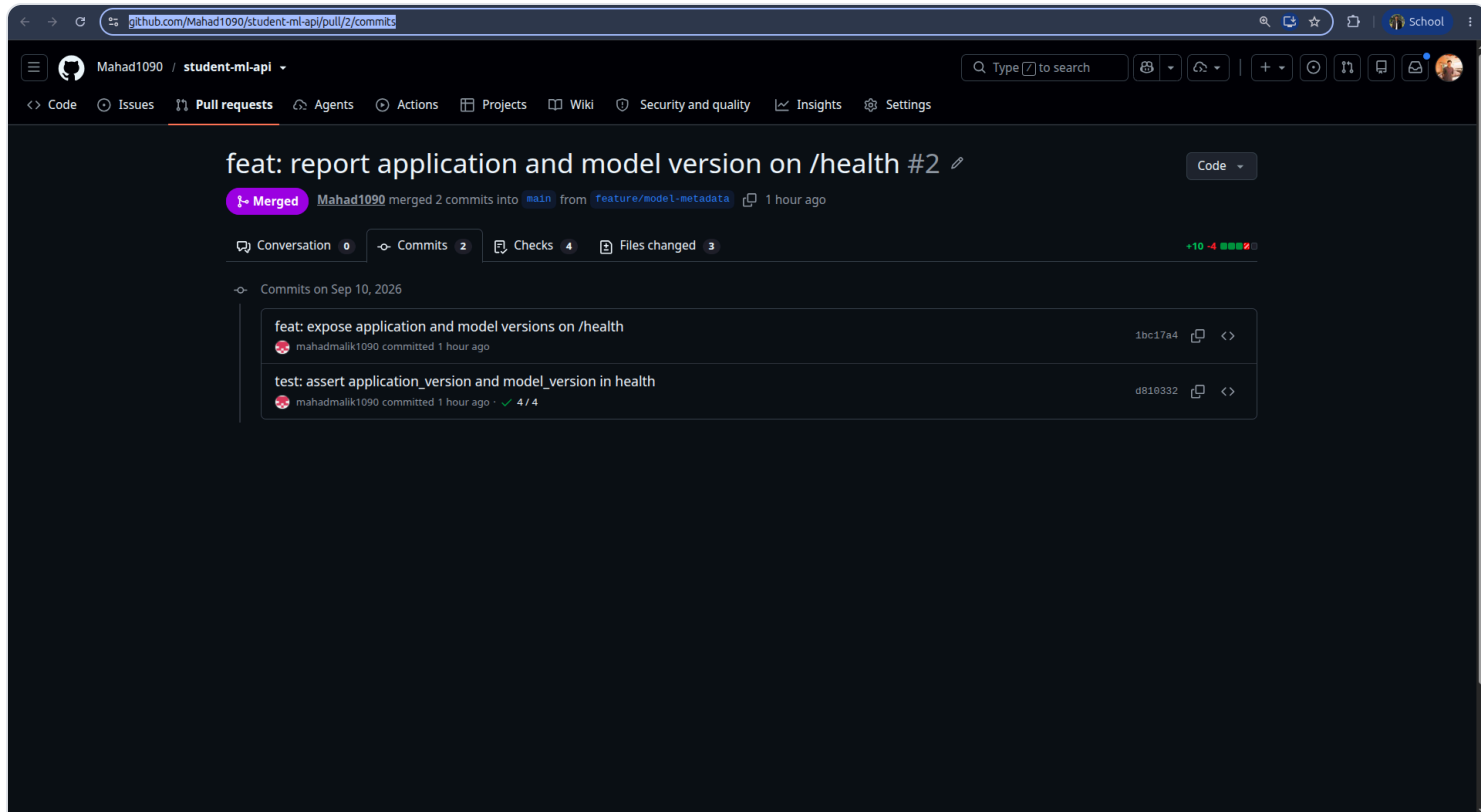


Fig 20 — PR #2 commits.

```
mahad1090@mahad-1090:~/Desktop/student-ml-api$ docker rm -f student-ml-api
docker pull ghcr.io/mahad1090/student-ml-api:1.1.0
docker pull ghcr.io/mahad1090/student-ml-api:latest
docker pull ghcr.io/mahad1090/student-ml-api:1.0.0
docker images --digests | grep student-ml-api      # latest & 1.1.0 share a digest; 1.0.0 differs

# run 1.1.0
docker run -d --name student-ml-api -p 5005:5000 ghcr.io/mahad1090/student-ml-api:1.1.0
sleep 1; curl -s localhost:5005/health; echo      # application_version 1.1.0, model_version model-1

# Part 20 - ROLLBACK: no source change, no rebuild, just run the old artifact
docker rm -f student-ml-api
docker run -d --name student-ml-api -p 5005:5000 ghcr.io/mahad1090/student-ml-api:1.0.0
sleep 1; curl -s localhost:5005/health; echo      # back to version 1.0.0

student-ml-api
1.1.0: Pulling from mahad1090/student-ml-api
6310eb16bf42: Already exists
dbd0b7849e6c: Already exists
1560cfed01dc: Already exists
56235e424563: Already exists
9056e4fd24fd: Pull complete
49bec192185d: Pull complete
6143975ac7a8: Pull complete
6fcb0731b20e: Pull complete
Digest: sha256:2c9ccbd13d38ad9e8fbb4a067825f32b2c39b09082b31e4372f03332baadcaf5
Status: Downloaded newer image for ghcr.io/mahad1090/student-ml-api:1.1.0
ghcr.io/mahad1090/student-ml-api:1.1.0
latest: Pulling from mahad1090/student-ml-api
Digest: sha256:2c9ccbd13d38ad9e8fbb4a067825f32b2c39b09082b31e4372f03332baadcaf5
Status: Downloaded newer image for ghcr.io/mahad1090/student-ml-api:latest
ghcr.io/mahad1090/student-ml-api:latest
1.0.0: Pulling from mahad1090/student-ml-api
Digest: sha256:10000c57bd3adc45a796199743b8b52e858b54d996ca34433f208c6b7b778e4c
Status: Image is up to date for ghcr.io/mahad1090/student-ml-api:1.0.0
ghcr.io/mahad1090/student-ml-api:1.0.0
ghcr.io/mahad1090/student-ml-api    1.1.0    sha256:2c9ccbd13d38ad9e8fbb4a067825f32b2c39b09082b31e4372f03332baadcaf5    5d19dd07f99e    3 minutes ago    179MB
ghcr.io/mahad1090/student-ml-api    latest    sha256:2c9ccbd13d38ad9e8fbb4a067825f32b2c39b09082b31e4372f03332baadcaf5    5d19dd07f99e    3 minutes ago    179MB
ghcr.io/mahad1090/student-ml-api    1.0.0    sha256:10000c57bd3adc45a796199743b8b52e858b54d996ca34433f208c6b7b778e4c    436f25f95b39    18 minutes ago    179MB
bc653d069cc0955cb23470b3a82c197409ed69e4783eb8cf779f8349c34de5f7
{"status":"healthy","application":"student-ml-api","application_version":"1.1.0","model_version":"model-1"}
student-ml-api
784860d7d340a1a6851052a82f37a5ba75300f7ddd4adf9798340e8f6eb89585
{"status":"healthy","application":"student-ml-api","version":"1.0.0"}
mahad1090@mahad-1090:~/Desktop/student-ml-api$
```

Fig 21 — digests: latest = 1.1.0 , 1.0.0 still present + rollback curls.

```

mahad1090@mahad-1090:~/Desktop/student-ml-api$ cd /tmp && rm -rf smoke && git clone https://github.com/Mahad1090/student-ml-api smoke && cd smoke

# inspect git history + PRs + tags
git log --oneline --decorate --all | head -15
git tag
git show v1.1.0 --stat | head -15

# pull the published artifact and run it
docker rm -f student-ml-api 2>/dev/null
docker pull ghcr.io/mahad1090/student-ml-api:1.1.0
docker run -d --name student-ml-api -p 5005:5000 ghcr.io/mahad1090/student-ml-api:1.1.0
sleep 1
curl -s localhost:5005/health; echo      # application_version 1.1.0, model_version model-1

# rollback to 1.0.0 - no rebuild, no code change
docker rm -f student-ml-api
docker run -d --name student-ml-api -p 5005:5000 ghcr.io/mahad1090/student-ml-api:1.0.0
sleep 1
curl -s localhost:5005/health; echo      # back to version 1.0.0
Cloning into 'smoke'...
remote: Enumerating objects: 29, done.
remote: Counting objects: 100% (29/29), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 29 (delta 4), reused 23 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (29/29), 17.23 KiB | 8.62 MiB/s, done.
Resolving deltas: 100% (4/4), done.
4650a2b (HEAD -> main, origin/main, origin/HEAD) docs: MLOps workflow write-up, traceability and failure analysis (#3)
02dc3db (tag: v1.1.0) feat: report application and model version on /health (#2)
fa0394b (tag: v1.0.0) feat: prediction API with CI pipeline (#1)
a8072f5 Initial commit
v1.0.0
v1.1.0
commit 02dc3db8bf199a185b6f0c9d0c454fb599e25e12
Author: Mahad Malik <168192717+Mahad1090@users.noreply.github.com>
Date: Thu Sep 10 20:41:24 2026 +0500

    feat: report application and model version on /health (#2)

    * feat: expose application and model versions on /health

    * test: assert application_version and model_version in health

-----

Co-authored-by: Mahad Malik <mahadmalik1090@gmail.com>
VERSION          | 2 +-
student-ml-api

```

Fig 22 — Demonstration: clone repo, inspect git history/tags.

```

sleep 1
curl -s localhost:5005/health; echo      # application_version 1.1.0, model_version model-1

# rollback to 1.0.0 - no rebuild, no code change
docker rm -f student-ml-api
docker run -d --name student-ml-api -p 5005:5000 ghcr.io/mahad1090/student-ml-api:1.0.0
sleep 1
curl -s localhost:5005/health; echo      # back to version 1.0.0
Cloning into 'smoke'...
remote: Enumerating objects: 29, done.
remote: Counting objects: 100% (29/29), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 29 (delta 4), reused 23 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (29/29), 17.23 KiB | 8.62 MiB/s, done.
Resolving deltas: 100% (4/4), done.
4650a2b (HEAD -> main, origin/main, origin/HEAD) docs: MLOps workflow write-up, traceability and failure analysis (#3)
02dc3db (tag: v1.1.0) feat: report application and model version on /health (#2)
fa0394b (tag: v1.0.0) feat: prediction API with CI pipeline (#1)
a8072f5 Initial commit
v1.0.0
v1.1.0
commit 02dc3db8bf199a185b6f0c9d0c454fb599e25e12
Author: Mahad Malik <168192717+Mahad1090@users.noreply.github.com>
Date: Thu Sep 10 20:41:24 2026 +0500

    feat: report application and model version on /health (#2)

    * feat: expose application and model versions on /health

    * test: assert application_version and model_version in health

-----

Co-authored-by: Mahad Malik <mahadmalik1090@gmail.com>
VERSION          | 2 +-
student-ml-api
1.1.0: Pulling from mahad1090/student-ml-api
Digest: sha256:2c9ccbd13d38ad9e8fbb4a067825f32b2c39b09082b31e4372f03332baadcaf5
Status: Image is up to date for ghcr.io/mahad1090/student-ml-api:1.1.0
ghcr.io/mahad1090/student-ml-api:1.1.0
7a8171e83e6f311109b1d802fe46f6bed47cf69f089683524e0441908da26807
{"status":"healthy","application":"student-ml-api","application_version":"1.1.0","model_version":"model-1"}
student-ml-api
ed02c9ea66f8a4219213a0ee337b26ef9a8d64c7ab1bd594fa2a59323c642775
mahad1090@mahad-1090:~/tmp/smoke$

```

Fig 23 — Demonstration: pull & run 1.1.0, then roll back to 1.0.0.

```

mahad1090@mahad-1090:/tmp/snoke$ gh pr view 2 --json number,title,state,mergedAt,mergeCommit,headRefName,baseRefName,url --jq '.'
gh pr list --state all
{
  "baseRefName": "main",
  "headRefName": "feature/model-metadata",
  "mergeCommit": {
    "oid": "02dc3db8bf199a185b6f0c9d0c454fb599e25e12"
  },
  "mergedAt": "2026-09-10T15:41:25Z",
  "number": 2,
  "state": "MERGED",
  "title": "feat: report application and model version on /health",
  "url": "https://github.com/Mahad1090/student-ml-api/pull/2"
}

Showing 3 of 3 pull requests in Mahad1090/student-ml-api that match your search

ID  TITLE                                                                                               BRANCH                                                                                               CREATED_AT
#3  docs: MLOps workflow write-up, traceability and failure analysis docs/mlops-workflow-writeup about 46 minutes ago
#2  feat: report application and model version on /health        feature/model-metadata    about 58 minutes ago
#1  feat: prediction API with CI pipeline                        feature/prediction-api    about 1 hour ago
mahad1090@mahad-1090:/tmp/snoke$

```

Fig 24 — Demonstration: inspect PR (`gh pr view` / `gh pr list`).

Workflow	Event	Status	Branch	Actor
docs: MLOps workflow write-up, traceability and failure analysis	CI #12: Pull request #3 opened by Mahad1090	Success	docs/mlops-workflow-writeup	Mahad1090
docs: record real run id for the missing-dependency failure	CI #11: Commit df9e5b7 pushed by Mahad1090	Success	docs/mlops-workflow-writeup	Mahad1090
chore: drop httpx to reproduce a missing-dependency CI failure	CI #10: Commit 498f2bc pushed by Mahad1090	Failure	chore/failure-demo	Mahad1090
feat: report application and model version on /health (#2)	Release #2: Commit 02dc3db pushed by Mahad1090	Success	v1.1.0	Mahad1090
feat: report application and model version on /health	CI #9: Pull request #2 opened by Mahad1090	Success	feature/model-metadata	Mahad1090
test: assert application_version and model_version in health	CI #8: Commit d810332 pushed by Mahad1090	Success	feature/model-metadata	Mahad1090
feat: prediction API with CI pipeline (#1)	Release #1: Commit fa0394b pushed by Mahad1090	Success	v1.0.0	Mahad1090
feat: prediction API with CI pipeline	CI #7: Pull request #1 synchronize by Mahad1090	Success	feature/prediction-api	Mahad1090
fix: correct health endpoint test		Success		Mahad1090

Fig 25 — Actions history (top): failed & successful runs.

Actions New workflow All workflows CI Release Management Caches Attestations Runners Usage metrics Performance metrics	✔ feat: report application and model version on /health (#2) Release #2: Commit 02dc3db pushed by Mahad1090 v1.1.0 Today at 8:42 PM 31s ...
	✔ feat: report application and model version on /health CI #9: Pull request #2 opened by Mahad1090 Feature/model-metadata Today at 8:40 PM 28s ...
	✔ test: assert application_version and model_version in health CI #8: Commit d810332 pushed by Mahad1090 Feature/model-metadata Today at 8:40 PM 29s ...
	✔ feat: prediction API with CI pipeline (#1) Release #1: Commit fa0394b pushed by Mahad1090 v1.0.0 Today at 8:27 PM 52s ...
	✔ feat: prediction API with CI pipeline CI #7: Pull request #1 synchronize by Mahad1090 Feature/prediction-api Today at 8:19 PM 38s ...
	✔ fix: correct health endpoint test CI #6: Commit 3f9d86f pushed by Mahad1090 Feature/prediction-api Today at 8:19 PM 38s ...
	✖ feat: prediction API with CI pipeline CI #5: Pull request #1 synchronize by Mahad1090 Feature/prediction-api Today at 8:17 PM 19s ...
	✖ test: break health assertion to demonstrate CI failure CI #4: Commit 952276f pushed by Mahad1090 Feature/prediction-api Today at 8:17 PM 14s ...
	✔ feat: prediction API with CI pipeline CI #3: Pull request #1 opened by Mahad1090 Feature/prediction-api Today at 8:15 PM 49s ...
	✔ ci: add tag-triggered release workflow and PR template CI #2: Commit 9b05928 pushed by Mahad1090 Feature/prediction-api Today at 7:52 PM 28s ...
	✔ ci: add pull request pipeline for tests and docker build CI #1: Commit 0e48625 pushed by Mahad1090 Feature/prediction-api Today at 7:50 PM 32s ...

Fig 26 — Actions history (bottom): Part-6 failed run, Release runs.

student-ml-api · Mahad Malik (231-0537) · Advanced MLOps Exercise · full write-up also in `docs/MLOPS_WORKFLOW.md` in the repository.